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TITLE : PPG CASE STUDY PIPELINE

PREPARED BY : GROUP 7

NAME	MATRIC NO
MUHAMMAD NAIM BIN ABDULLAH	A23CS0134
MUHAMMAD SYAHMI FARIS BIN RUSLI	A23CS0138
JOANNE CHING YIN XUAN	A23CS0227
SABRINA HENG WEI QI	A23CS0265
NURUL ADRIANA BINTI KAMAL JEFRI	A23CS0258
JEANETTE HAUW CHANDRA	X25EC3020
DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220

PREPARED FOR: DR. ARYATI BINTI BAKRI

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ABSTRACT

In industrial companies, inventory management is essential since poor inventory control can result in excess inventory, dead stock, expired supplies, stockout concerns and large financial losses. PPG Coatings (Malaysia) Sdn. Bhd. struggles to manage inventory data that is dispersed over several operational datasets, which limits inventory visibility and makes it harder to spot possible inventory hazards. In order to improve inventory monitoring and facilitate data-driven decision-making, this project intends to provide a cloud-based data engineering pipeline for Recoverable Assets (RA) and Inventory Risk Management (IRM). The suggested solution uses the System Development Life Cycle (SDLC) methodology and the Bronze, Silver and Gold LAyers of the Medallion Architecture. Microsoft Azure technologies were used to ingest, integrate, clean, process and analyze six operational datasets. Recoverable Assets, Magna Carta Dead Stock, Magna Carta Expired Materials, stockout risks, financial provisioning requirements and impacted client sales orders were all identified using business rules. After processing, the data was organized into a star schema and shown using interactive Microsoft Power BI Dashboards. Every component of the proposed pipeline functioned efficiently and satisfied the system requirements, according to the results of testing and validation. The research found 17 recoverable assets, one Magna Carta Dead Stock item, two expired supplies, a total financial provision amount of RM147,565,00.00 and two customer sales orders affected by stockout risks. The proposed system enhances inventory visibility by integrating heterogeneous data into a unified platform and providing automated inventory risk assessment and reporting functionalities. The research concludes by showing how contemporary data engineering and business intelligence technologies can enhance operational decision-making in manufacturing settings, lower inventory-related risks and enable proactive inventory management.

Keywords: Inventory Risk Management, Recoverable Assets, Data Engineering Pipeline, Medallion Architecture, Business Intelligence.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Storing excess recoverable assets, especially when raw material stocks far exceed actual production demands, introduces severe risks to a manufacturer's financial health. [Unverified] If left unmanaged, these surplus materials degrade into expired or obsolete dead stock, fulfilling the Magna Carta conditions for inventory obsolescence (Yerra, 2025). Lacking automated monitoring systems, companies struggle to accurately measure these inventory-based financial risks, leading to artificially inflated asset valuations and the squandering of warehouse capacity (Holloway, 2024).

Furthermore, a substantial gap frequently emerges between stored inventory and active order fulfillment. Because of unpredictable market demand and fluctuating supplier lead times, businesses might harbor massive surpluses of certain materials while facing critical shortages of others (Heizer et al., 2020). These operational bottlenecks are typically rooted in fragmented data architectures that prevent real-time transparency and severely restrict strategic planning (Holloway, 2024; Heizer et al., 2020).

Inadequate data visibility consistently jeopardizes inventory risk management, occasionally resulting in catastrophic monetary losses. The historic Cisco Systems inventory crisis illustrates this perfectly; an inability to differentiate viable stock from obsolete components during market turbulence caused a write-down of approximately USD 2.25 billion. This incident highlights the critical need for reliable, real-time inventory tracking to guide corporate strategy (Fixinc, 2025). Ultimately, basic carrying costs consume 20% to 30% of total inventory value, and poor system integration only accelerates the buildup of worthless stock, continually degrading operational output (Russell & Taylor, 2019).

1.2 Problem Background

In contemporary manufacturing, the growing stockpile of surplus raw materials poses a significant operational hurdle. Without adequate management, materials stored beyond anticipated production needs are highly prone to becoming dead stock or aging into obsolescence. This problem is heavily compounded by a lack of inventory transparency, leaving businesses without up-to-the-minute data regarding stock health, consumption trends, and product lifecycles. Consequently, organizations struggle to pinpoint excess goods or flag materials nearing obsolescence, which ultimately drives up operating expenses and wastes valuable resources (Holloway, 2024; Heizer et al., 2020).

Furthermore, relying on disconnected data systems generates a fractured information landscape that stops companies from gaining a comprehensive, supply-chain-wide perspective on their inventory. Lacking a centralized, automated flow of data, inventory records become siloed, severely restricting a firm's capacity to synchronize purchasing, manufacturing, and demand forecasting. This lack of integration sharply elevates financial exposure, as surplus and outdated stock can easily go unnoticed until expensive inventory write-downs are completely unavoidable (Russell & Taylor, 2019; Holloway, 2024).

1.3 Project Aim

The aims of this project are to design and develop an end-to-end data engineering pipeline for inventory risk management, focusing on the identification and monitoring of recoverable assets and inventory. The system aims to automatically detect and track raw materials, classify items at risk of becoming dead stock, and quantify potential financial exposure associated with excess and obsolete inventory.

Additionally, this project seeks to enhance inventory visibility by integrating data sources into a unified, analytics-ready data model. Through the implementation of real-time dashboards and key performance indicators, the proposed solution to support data-driven decision-making, reduce inventory-related risks, and improve the alignment between supply, demand, and financial reporting within manufacturing operations.

1.4 Project Objectives

A number of objectives have been identified to direct the creation of the suggested Recoverable Assets (RA) and Inventory Risk Management(IRM) system based one the project's objectives. The following are the objectives:

1. To offer end-to-end data engineering pipelines for inventory risk management using a layered architecture (Bronze, Silver, Gold Layers).
2. To automatically recognize and classify inventory risk, such as stockout hazards, dead stocks, expired supplies and Recoverable Assets (RA).
3. To integrate multiple scattered data sources into a single data platform in order to improve data consistency and visibility.
4. To implement ETL/ELT processes to ensure that data is properly cleaned, transformed and stored for analysis.
5. To use business information tools to construct dynamic dashboards that enable real-time monitoring and decision-making.
6. To determine the potential financial risk associated with access and obsolete inventory.

1.5 Project Scope

The primary goal of this project is to establish a data-driven inventory risk management system in a manufacturing scenario. The scope of the project includes:

1. Data Integration
    - Assembling and merging inventory data from several sources into a common storage system.
2. Data Engineering Pipeline
    - Employing a three layers structure (Bronze, Silver and Gold Layer) to handle unrefined, processed and analytically prepared data.
3. Data Processing (ETL/ELT)
    - Ensuring data quality and usefulness for analysis by carrying out data extraction, transformation and loading processes.
4. Inventory Risk Identification
    - Identifying stockout problems, obsolete inventory, expired items and Recoverable Assets (RA) based on inventory state by using business techniques.
5. Cloud-based Implementation
    - Utilizing Microsoft Azure for data storage, processing and management.
6. Data Visualization
    - Creating Power BI dashboards to provide crucial insights and aid in decision making.
7. limitations
    - Machine learning models and sophisticated predictive analytics are not included in this study which mainly focuses on rule-based risk identification.

1.6 Project Importance

This project is important because it addresses important problems with inventory control and financial risk in the modern industrial system. By combining disparate data into a single system, the project improves inventory visibility and helps businesses better manage stock levels and conditions. Second, proactive risk management is made possible by identifying dead stock, excess inventory and expired goods before they result in significant financial losses. Additionally, compared to traditional inventory management methods using an automated data engineering pipeline reduces human error and improves data accuracy. Furthermore, by presenting insights via interactive dashboards and key performance measures (KPIs), data visualization technologies enhance data-informed decision-making. Lastly, by balancing supply and demand and improving stock management, this project also helps reduce inventory costs and improve operational efficiency which eventually improves overall business performance.

1.7 Report Organization

This report is structured into 5 chapters to provide an overview of this project

Chapter 1 : Introduction

This chapter provides the background of the project, along with the project's overall aim, specific objective, scope, and importance to modern manufacturing

Chapter 2 : Literature Review

This chapter discuss the fundamental theories of data engineering pipelines and inventory management concepts, reviews related previous research, and the details of the core technologies that will be utilized, such as Microsoft Azure and Power BI

Chapter 3 : Methodology

This Chapter Discusses the chosen project methodology, outline the specific phases of development, and justifies the technology stack used to build the solution

Chapter 4 : Result and discussion

This chapter present the details the system architecture, requirement analysis, database design, and interface design required to structure the three-layer data pipeline and visualization dashboard

Chapter 5 : Conclusion and Recommendation

This chapter summarizes the overall project outcomes, evaluates the successful achievement of the project objectives, and provides recommendations for future improvements.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter introduces the core concepts, systems, and technologies that underpin the proposed Recyclable Assets (RA) and Inventory Risk Management (IRM) system. It develops a theoretical background for the conceptual foundation and system development of the proposed RA and IRM system by discussing supply chain management, Enterprise Resource Planning (ERP) systems, Medallion architecture-based data engineering pipelines, ETL/ELT processes, inventory management principles, and data visualization,

In addition, this chapter discusses related systems and research on inventory analysis and data-driven solutions. Through this comparison, this chapter points out the advantages, disadvantages, and weak points of these systems, and current research results reveal gaps and the necessity for the proposed system. Additionally, the chapter also demonstrates how the technologies, such as Microsoft Azure and Power BI, facilitate the conceptual development of the proposed IRM system through efficient data processing, storage, and visualization.

Overall, this literature review forms the theory and technological foundation needed to design and develop the proposed inventory risk management system.

2.2 Fundamental Theory & Concept

This section explains the fundamental theories and concepts related to data engineering, inventory management and data visualization of the proposed system.

2.2.1 Data Engineering Pipeline

A data engineering pipeline refers to a series of processes used to collect, transform and store data for analysis. It ensures that raw data is converted into meaningful and usable information for decision-making. In this project, a three-layer data pipeline is used:

- Bronze Layer (Raw Data)

  This layer stores raw data from multiple sources into a data lake. Basic data quality checks such as null detection, duplicate data removal and data type validation are performed.

- Silver Layer (Processed Data)

  Data is cleaned and transformed in this layer. Business logic starts to be applied to identify inventory conditions such as excess stock, dead stock and expired materials.

- Gold Layer (Analytics- Ready Data)

  This layer contains structured and optimized data, typically organized into data warehouse schema. It is used for reporting and visualization through tools like Power BI.

The layered data pipeline improves data quality, scalability and maintainability of the system.

2.2.2 ETL/ELT Process

ETL means Extract, Transform, Load, which is a process used in data integration. Extract is the process of collecting data from various sources such as CSV files. Transform is a process of cleaning, filtering and process according to business rules. Load means transformed data is stored in a data warehouse for analysis. The ELT approach is often used in modern cloud systems such as Microsoft Azure, in which data is loaded into storage and then transformed. This helps to enhance performance and scalability.

2.2.3 Inventory Management Concepts

Inventory management is a process of keeping track of and managing inventory to ensure it is used efficiently and losses are minimized. The following are the key concepts that were used in this project:

- Recoverable Assets (RA)

  These are materials with stocks of more than 12 months of consumption. This means that there is overstocking and potential financial inefficiency.

- Magna Carta Dead Stock

  Materials that are not used over 9 months and have no expiry date. These will not be used and can lead to financial loss.

- Expired Materials

  Materials that are no longer usable due to their expiration date. These require full financial write-down.

- Stockout Risk

  This happens when stock is depleted and demand is not being met, particularly due to the time it takes for suppliers to deliver. It may result in delayed production or unfulfilled customer orders.

These ideas are crucial in risk assessment and inventory decision making.

2.2.4 Data Visualization and Business Intelligence

Data visualization is the process of displaying data in graphical formats like charts and dashboards. It facilitates the understanding of patterns, trends and insights more effectively. Monitoring inventory levels, identifying stock-related risks such as overstock and stockouts, and providing insights for decision-making with interactive dashboards are all facilitated by Business Intelligence tools like Microsoft Power BI. Visualization is a key element in the process of turning transformed data into actionable insights.

2.3 Review of Related Research and Existing Systems

Today's industrial world has made inventory management a strategic component of supply chain resiliency. In the past, companies focused their management efforts on trying to balance replenishment lead times (RLT) and carrying costs, based on fixed mathematical models. But with the advent of the disruptions in the global market, the need has been created to shift to the "intelligent" data-driven ways of doing things. For large-scale operations, the process of accurately identifying material obsolescences becomes a key part of ensuring profitability and efficiency, with the risk of substantial financial write-downs when these are not identified.

Past sales records are a "critical element" for forecasting demand and reducing stockouts, but they should not be used alone as it leads to inaccuracy in forecasting (Liu et al., 2025). They highlight that today's systems have to combine several factors, such as having access to the current state of inventory and short-term demand predictions, to deal with the challenges of big data, which is frequently “imbalanced”, meaning that critical risk events are rare but have a great effect. It confirms the need for the “Silver Layer” in the proposed data pipeline that will not only track sales but also employ sophisticated business rules to identify “Recoverable Assets”.

In addition, the conventional methods of inventory which are largely based on sampling and manual counting are; labor intensive, time consuming, and prone to human error (Mulaka et al., 2025). They have also worked on the topic of automated forest nursery systems, showing how the transition towards real-time, automated inventory systems can lead to better reliable data, directly improving decision-making, while also reducing operating costs. With thousands of lot expiry dates, and millions of data rows, this automation is crucial for an industry leader in chemical manufacturing to avoid human error when tracking these.

Finally, the found obsolescence is actually a major life cycle cost driver which can cost the organization up to 5% to 40% more than the optimum amount on the economic consequences of stockout (Gravier and Swartz, 2009). It is firmly believed that proactive obsolescence management strategies over reactive strategies are needed in relation to holding dated stock with the aim of reducing the amount of large expenses.

The proposed Azure pipeline needs to be compared with the existing system to strengthen the proposed pipeline. **Table 2.3.1** gives an overview of the current inventory management systems along with the proposed cloud-based inventory management system. This table compares manual processes and old-style Enterprise Resource Planning (ERP) reporting with the new data engineering, focusing on data integrity, complexity of logic, operational speed and financial risk.

Table 2.3.1 Comparison of Existing Systems

Feature	**Traditional Manual (Excel)**	**Legacy ERP Reporting (SAP/Oracle)**	**Proposed Azure Pipeline**
Data Integrity	High risk of human error (Mulaka et al., 2025).	Moderate; often siloed data (Gorelik, 2019).	High; automated ingestion and cleaning (armbrust et al., 2021).
Logic Complexity	Limited to basic formulas.	Rigid; hard to customize for specific rules.	Flexible; applies complex Magna Carta logic.
Operational Speed	Reactive; slow updates.	Periodic batch updates.	Proactive; real-time analytics ready.
Financial Risk	High; 40% cost increases risk (Gravier & Swartz, 2009).	Low visibility into dead stock.	Low; automates provision calculations.

To summarize, literature shows that traditional inventory systems are in most cases reactive, meaning that they only detect obsolescence and/or stockouts when they affect the bottom line. Manual processes have been found by previous research to be prone to error and sales-based approach to forecasting is not adequate for modern risk management. The proposed three-layer Azure pipeline, on the other hand, is the proactive management approach suggested by industry experts. It combines automation to minimise human error with data intelligence to support the big stakes financial risks of material expiry and the "Magna Carta" dead stock policy.

2.4 Technology Used

When it comes to building a pipeline, there are several different tools that can be used to get everything work.

2.4.1 Microsoft Azure

This are the services that the Microsoft Azure have :

Azure Data Factory

Azure Data Factory (ADF) acts as an automated delivery system and traffic controller for your entire project. ADF used automated workflows called “pipelines” to extract the data from the resources and load it into the storage system (Microsoft, n.d.-a). ADF does not alter, clean or analyse the data itself. Its responsibility is to make sure the raw data is transported on schedule into the bronze layer so the rest of the data pipeline can begin its work (Microsoft, n.d.-b).

Azure Data Lake Storage

Azure Data Lake Storage is a massive and secure place that you can load your files. It's like a highly scalable storage that is designed to hold a massive amount of big data in its original format (Microsoft, n.d.-c) . In this project, this will act as a Bronze Layer because it allows storing all of the raw CSV files securely. It acts as the foundation storage layer before any complex cleaning or processing start (Microsoft, n.d.-d)

Azure SQL Database

Azure SQL Database is needed to create perfectly structured data so that business reports can read it easily and quickly. This fully managed relational database acts as Gold layer (Microsoft, n.d.-g) . These tools will organize the refined data into formatted “Star Schema,” which separates the data into fact tables and dimension tables. Instead of a vast lake of loose files, the SQL Database provides a highly organized, indexed, and instantly searchable environment that is efficient for analytics (Microsoft, n.d.-h).

2.4.2 Microsoft Power BI

Microsoft Power BI serves as the visual representation layer that brings together all of your data work to life for business stakeholders (Microsoft, n.d.-i). Business managers do not want to look at the multiple rows of numbers in a database, instead they want to visualise the data where Power BI comes handy because it can translate those numbers into an interactive dashboard (Microsoft, n.d.-j). It allows us to build the specific graft, chart, and KPIs that are required to achieve the project goals (Microsoft, n.d.-k).

2.5 Summary

This chapter provided an overall review of the required essential knowledge, concepts, theories, and techniques to build the Recyclable Asset (RA) and Inventory Risk Management (IRM) system. It also highlighted the dependency of the solutions on the integration of SCM and ERP in addition to up-to-date Data Engineering principles like the Medallion architecture.

The chapter also illustrated the use of data engineering pipelines and ETL/ELT techniques for transforming unprocessed data into trusted and usable information. Besides, it presented the inventory management concepts, including RA, Magna dead stock, expired materials, and stock-out risks, by showing the ways in which data-driven solutions would help address the inefficiency and financial loss.

Furthermore, the review on available systems and research also uncovered the restrictions of current inventory management methodologies, such as overdependence on historical sales records may lead to PFA workload and inaccurate sales forecasts, manual inventory management may be time-consuming and error-prone, and incorrect outdated inventory management may result in heavy losses. More importantly, the Data Engineering technologies like Microsoft Azure and Power BI as introduced in the chapter pave the foundation and provide industry-strength infrastructure and outstanding development tools to support the modern end-to-end data pipeline deployment. The section has successfully strengthened our ability of data integration, process, and visualisation.

Overall, the key strengths of the chapter had contributed greatly to project planning and subsequent system design in later chapters.

CHAPTER 3: METHODOLOGY

3.1 Introduction

The approach used to create the suggested Recoverable Assets (RA) and Inventory Risk Management (IRM) system is presented in this chapter. While the preceding chapter covered the theoretical principles, relevant works, and technologies involved, this chapter concentrates on the strategy and techniques employed throughout the system's development in an organized manner.

The methodology acts as a framework to ensure that the development process is carried out consistently, methodically and in line with the project's objectives. The System Development Life Cycle (SDLC) approach which guides the planning, system design, implementation, testing and evaluation activities at each stage, is used to build the full data engineering pipeline for this project. Every stage of the procedures is carefully planned and adjusted to meet the needs of data processing, data integration and data analysis or inventory risk management.

Additionally, this chapter describes each stage of the development process, describes the technologies that help with system implementation and provides justification for the approach that was selected. Through a methodical data engineering strategy to improve inventory risk assessment and decision-,making, the project aims to guarantee dependability, efficiency and expansion.

3.2 Methodology Choice and Justification

The main approach used in this research is Medallion Architecture, which is a data design pattern that is used to structure data into a sequence of functional layers in a data lakehouse environment (Armbust et al., 2021). This architecture is based on a “multi-hop” process to progressively enhance the quality and structure of data as it flows through the process (Microsoft, 2024). Other architecture like Lambda Architecture –where there are two processing streams for batch and real-time data were explored, but were explored, but were not found to be a viable option for this project. Since PPG’s data refresh cycle is batch-oriented, they did not need to have high-frequency real-time stream processing layer; therefore, they opted for the Medallion approach for its simplicity of maintenance and greater data consistency (Microsoft, 2024).

The first reason for this choice is to avoid the data swamp. A data swamp is an unmanaged unstructured data lake that doesn't have sufficient metadata, governance, data quality control, or trust, making the data stored in the lake inaccessible or unreliable for business decisions (Gorelik, 2019). The group's three-layer architecture, which are Bronze, Silver and Gold –transitions data from raw ingestion to analytics-ready insights, and continues to provide data engineers full auditability throughout the data engineering lifecycle (Armbrust et al., 2021), as shown in the **Figure 3.2.1** below.

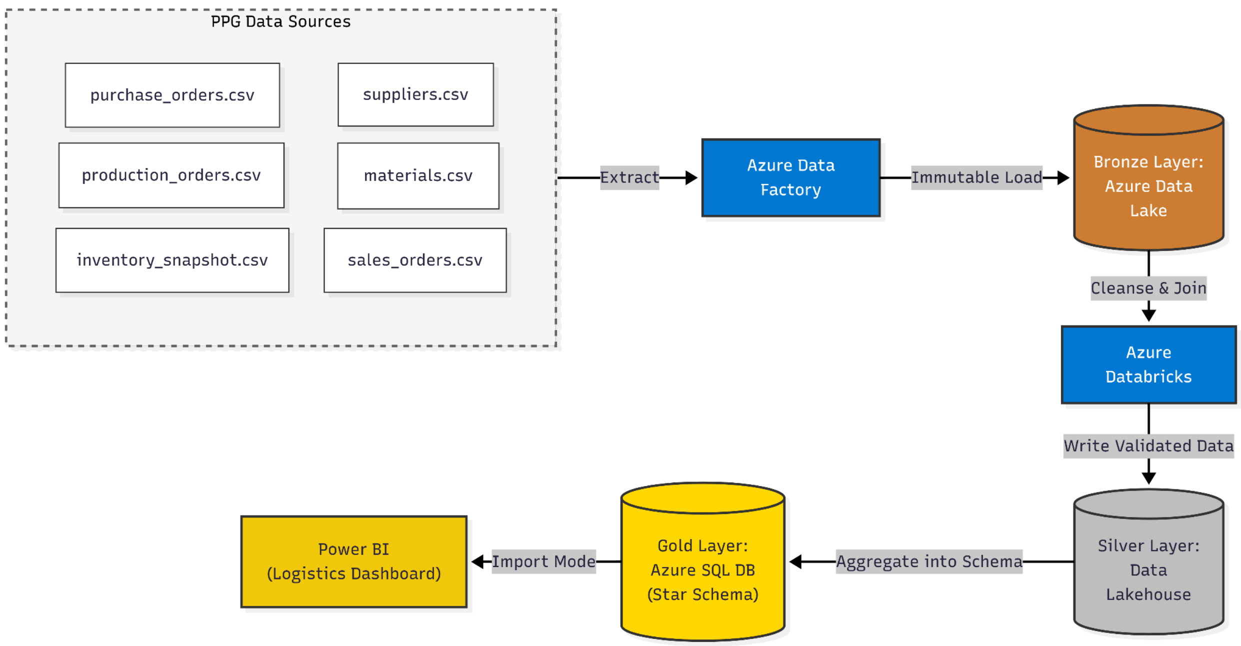


Figure 3.2.1 *Overview of the Medallion Architecture*

3.2.1 Hierarchical Data Refinement Layers

The data engineering process is broken into three stages of refinement: a first stage for refining the data, a second stage for defining the data architecture, and a third stage for implementing the Medallion Architecture. This is a hierarchical approach with the idea that the pipeline would be separated by the maturity of the data. The next layer performs a different function than the previous one, ranging from preserving raw data to a quality control step, and then a step of analytical structuring of the data stage that runs well. There are three different layers (Bronze, Silver, and Gold) to the processing activities, each with a specific technical justification.

3.2.1.1 Bronze Layer (The Auditability Argument)

Raw data that is uploaded from the six PPG CSV files (such as materials.csv or purchase_orders.csv) will land in the Bronze layer. To keep the pipeline "non-destructive", data is stored in its native format (Microsoft, 2024). Financial accountability is a major driver in the life cycle of this layer, so the immutability of inventory snapshots is crucial for audit and recordkeeping –justification for this layer is auditability (Gravier & Swartz, 2009).

The key for preserving historical data for obsolescence risk management directly maps into the non-destructive data ingestion approach of the Bronze layer. It can offer PPG with a history of the baseline and the ability to recalculate the results from the source if the results are adjusted according to the business rules in Magna Carta (Gravier & Swartz, 2009). This layer adds storage overhead because of data redundancy, but is a trade-off for ensuring that the audit is accurate.

3.2.1.2 Silver Layer (The Quality Control Argument)

Data is cleaned, integrated and validated in the Silver Layer. At this point, the PPG business rules kick-in, which is the calculation of Recoverable Assets (RA) and flagging Dead Stock. This layer is used to isolate for incremental quality refinement. The Silver Layer is a cleaning station, where data from different sources is integrated and transformed into business schemas (Arnbrust et al., 2021).

This is better than classic single-step ETL processes, as it enables fine-grained validations (like checking for nulls or duplicates in inventory_snapshot.csv) to be addressed before more complex business rules are performed. The complexity of the process to manage these transformations, however, makes the processing time of the pipeline longer, which was addressed by using automatic orchestration using Azure Data Factory.

3.2.1.3 Gold Layer (The Performance Argument)

The Gold layer is the most refined layer in which data is presented in a star schema for visualization. This is a fact_inventory_status table that is surrounded by descriptive dimension tables such as dim_material and dim_supplier. A star schema is appropriate when having high performance for queries is warranted. Dimensional modeling has been designed to favor data retrieval speed for analytical tools, such as Power BI by avoiding complex table joins, according to Kimball methodology.

This directly aligns with the project's objective to provide the answers to the seven business questions that PPG has outlined; for example, it helps identify which sales orders are impacted by a shortage and are required to be filled within a few days. The Gold layer is more rigid than the Bronze layer when it comes to how easily raw data can be explored, however, its structured approach allows for logistics managers to gain quick, consistent and actionable insights.

3.3 Phases of the Chosen Methodology

3.3.1 Process and Activities within Each SDLC Phase

This section highlights the procedure and activities of all the phases of the SDLC. While this activity had the primary structure of five phases, adjustments have been made to comply with the specific development process for the Recyclable Assets (RA) and Inventory Risk Management (IRM) systems in this project. These phases will lead to the systematic building of the system and will be useful for the data engineering pipeline to be successfully implemented. These stages are extensible and adaptable to smoothly process and analyse data.

Phase 1: Planning and Requirement Analysis

During this stage, the aim and scope of the project were clarified. Through the case studies from PPG and the six source data sets, the inventory position and major issue of business could be explored. Areas such as overstock, slow-moving goods, expired materials, and out-of-stock risks were all analysed as they would have an impact on production and customer orders.

The system was designed to classify by business requirements suggested in the case study, like Magna Carta (MC) Dead Stock and Expired Materials, stockout risk and impact on production and sales activities, and examined the out-of-stock risk and effect on production and sales business according to the respective business needs in the case study. These requirements were transformed into seven business questions (Q1 –Q7) used to develop the final solution.

To perform the analysis, six data sets including materials.csv, inventory_snapshot.csv, production_orders.csv, purchase_orders.csv, suppliers.csv, and sales_orders.csv, were all analyzed and merged on the common attribute material_id. Based on the established performance and business standards, business rules and performance metrics were established for inventory risk management for the final process.

Phase 2: System Design

In the system design stage, the business requirements established in the first stage are transformed into a hierarchical database scheme & analysis model. The usage of finding RA, MC Dead Stock, Expired Materials, Stock Out orders risks, and their influence on production & customer orders are converted into data calculation rules, table structure, and reports.

Medallion Architecture was designed using the Bronze, Silver, and Gold Layers. The original six existing data sets are all stored in the bronze layer in their original state. Silver layer is used to perform data cleaning, integration, and standardisation across multiple data sets. material_id exists in all data sets and is used as the key attribute across the data sets. The objective of the gold layer is to perform the undertaking of logic, or business rule application, and perform calculations to meet the business requirements. For example, RA classification, MC obsolete inventory identification, expired material detection, reserve amount calculation, out-of-stock risk analysis, and the sales impact analysis

Moreover, a Star Schema structure is established to enable setup analysis reports and create dashboards. The central fact table is fact_inventory_status, capturing inventory-related indicators and other business data. The descriptive details are contained in dimension tables, such as dim_material, dim_supplier, and dim_time. This architecture allows the process of querying data, tracking key performance indicators, and visualizing data in the Power BI dashboard to be carried out efficiently.

Phase 3: Development (Data Engineering Lifecycle Implementation)

The development of stage involves the creation of a data pipeline using the services provided by Microsoft Azure. Once the life cycle of data engineering has completed, the CSV data set are imported into Azure Data Factory and stored in Azure Storage. A quality check, including empty value check, duplicate check, and data type check, is carried out.

Next, 3 layers architecture Bronze, Silver and Gold are used across Azure platform to process and transform data, respectively. Silver layer mainly performs data cleaning, integration and standardisation tasks, while Gold layer takes care of implementing business rule processing tasks and performing analysis calculations, including the 12-month consumption analysis, recyclable assets (RA) classification, identifies dead and expired materials according to the Magna Carta (MC), accrual calculation, out-of-stock risk analysis and out-of-stock-order impact analysis.

Finally, the processed data was put into a Star Schema structure of fact tables and dimension tables for reporting and analysis. Power BI dashboard was also created to visualize inventory risks, material status, configuration volume, and out-of-stock consequences to help customers make business decisions effectively.

Phase 4: Testing

The testing phase is the one in which it engages in a sequence of comprehensive checks to ensure that the technical setup and all the formulas function flawlessly. The first activities were to check the correctness of basic calculations, focus for RA classification, expiry detection and provisioning. In addition, the entire data pipeline was run on the bronze, Silver, and Gold layer to ensure a seamless transfer of data from the local server to cloud without any errors or missing values. Last but not least, the output on the Power BI dashboard will be compared to the anticipated record to guarantee that visuals on the report are accurate. This phase was the last quality assurance that no problems at all would occur when the system was put into production.

Phase 5: Deployment and Evaluation

In this phase, the system will be published on the Microsoft Azure platform. After the system was in operation, a comprehensive evaluation was made to make sure that it met the company's requirements. The review validated that data remained accurate and consistent during the process. This was the last step to make sure that the finished pipeline meets the business requirements.

3.3.2 Design Modelling

Data Flow Diagram (DFD)

Figure 3.3.2.1 illustrates the Data Flow Diagram on how raw CSV data is ingested through Azure Data Factory into Azure Data Lake, processed using Azure Databricks, and stored in Azure SQL Database for analytical reporting in Power BI.

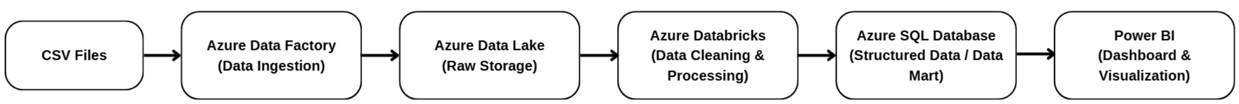


Figure 3.3.2.1: Data Flow Diagram (DFD) of the Proposed Azure Data Pipeline

System Architecture Diagram

Figure 3.3.2.2 shows the system architecture that integrates Azure services to support scalable data processing. Azure Data Factory handles data ingestion, Data Lake stores raw and processed data, Databricks performs transformation and analytics, SQL Database stores structured data, and Power BI provides visualization.

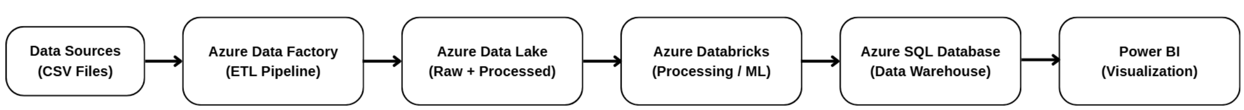


Figure 3.3.2.2: System Architecture of the Azure-Based Inventory Risk Management Pipeline

Star Schema Diagram

Figure 3.3.2.3 illustrates the star schema that organizes data into fact tables capturing transactional events and dimension tables providing descriptive context, enabling efficient querying and analysis for inventory risk management.

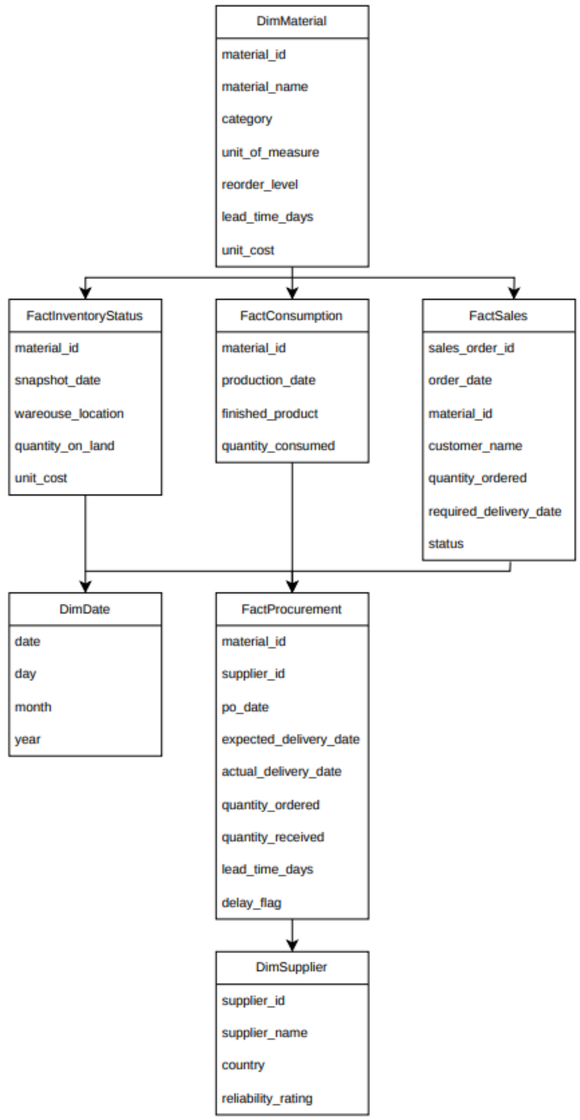


Figure 3.3.2.3: Star Schema Design for Inventory Risk Management Analytics

3.3.3 Design Tools

The design of the Recoverable Assets)RA) and Inventory Risk Management (IRM) system requires clear diagrams to explain the system structure, data movement, and database design. To support this process, several design tools were used to ensure the system is organised and easy to understand.

Draw.io and Lucidchart were used to create the system architecture diagram and Data Flow Diagram (DFD). These diagrams show how data moves through the Bronze, Silver, and Gold layers of the data pipeline. They also illustrate the conncetion between Azure data Factory, Azure Data Lake Storage Gen2, Azure Synapse Analytics (Serverless SQL), and Power BI. This helps users undeerstand how the different components work together within the systems.

In addition, Power BI Model View was used to design and verify the star schema in the Gold Layer. It was used to define the relationships between the fact table, fact_inventory_status, and the dimension tables, including dim_material, dim_time, and dim_supplier. This ensures that the data model is accurate and suitable for reporting and analysis.

Overall, these design tools helped improve the clarity of the system design and ensured that the system structure met the project requirements.

3.4 Technology Used Description

This project makes use of multiple Azure tools along with Power BI for designing an end-to-end data engineering pipeline for inventory risk management. The selection of technologies depends on their role in each phase of the data engineering pipeline, such as data ingestion, data storage, data processing and data visualization.

3.4.1 Azure Data Factory

Azure Data Factory is used for managing and automating the data ingestion process. Specifically, in this project, Azure Data Factory is used to extract data from multiple CSV files and load data into the storage layer. The use of Azure Data Factory pipelines allows automating the movement of data and makes the process more reliable and efficient.

3.4.2 Azure Data Lake Storage

Azure Data Lake Storage is used as a main storage layer for all raw data and constitutes a Bronze layer of the data pipeline. The Bronze layer consists of storing data in its initial form before data processing. The reason for using Azure Data Lake Storage is scalability and ability to store large amounts of data. Storing data in the initial form in this layer is helpful for possible future processing of data and allows tracing the origin of data.

3.4.3 Azure SQL Database

The next layer, the Gold Layer, is Azure SQL Database, where the processed data is saved in structured form. Azure SQL database is designed to organize data according to the star schema, involving fact and dimension tables. It is used for conducting analytical processes and provides the best performance.

3.4.4 Microsoft Power BI

The last technology used in this project is Microsoft Power BI. It is used for visualizing the processed data in a form of interactive dashboards. Power BI works directly with Azure SQL database to show the processed data through various types of charts and KPIs.

3.5 System Requirement Analysis

In this section, the system requirements for the Inventory Risk Management system will be discussed. These requirements can be subdivided into functional and non- functional aspects.

3.5.1 Functional Requirements

Here are some functional requirements that the system should meet:

- Data Ingestion

  It is necessary to import data from various CSV files, which include such categories of information as material, inventory, production, supplier, purchase order and sales order into the data storage subsystem.

- Data Cleaning and Validation

Data quality control should be performed, including data normalization in terms of dealing with missing data, deleting duplicates and checking data types before processing.

- Data Transformation and Processing

  To obtain relevant results, certain actions should be performed by the system such as identifying Recoverable Assets (RA) depending on their consumption, detecting Magna Carta Dead Stock, identifying expired materials and also calculating stockout risks.

- Data Integration

  It is required to unify data from different sources into a single dataset that could show the state of inventory at a particular moment.

- Data Storage

  To enable data analysis later on, it is needed to organize data using the data warehouse strategy.

- Data Visualization

  To provide users with opportunities to analyze data and manage inventory risks, it is necessary to build interactive dashboards.

3.5.2 Non-Functional Requirements

Non-functional requirements are the following:

- Scalability

  With growing data, it is important that the performance of the system remains stable.

- Data Quality and Reliability

  It is necessary to guarantee data accuracy, consistency and integrity.

- Performance

  The system should perform tasks quickly, giving the opportunity to receive results in a timely manner.

- Security

  The data storage and analysis should be performed in a protected way preventing any unauthorized actions.

- Usability

  Users' dashboards should provide an easy way to analyze data and make conclusions.

- Maintainability

  System modification should be easy, especially in case some changes are made to data sources or business logic.

3.6 Gantt Chart

The project's Gantt chart is in **Figure 3.6.1**, which is the timeline for the development of the PPG inventory pipeline from late March through late June 2026, a period of 13 weeks. The schedule has four main phases: Introduction (Requirements and scoping), Data Model & Pre-Processing (Data collection and data schema design), Dashboard Implementation (Data cleaning, transformation, storage and PowerBI visualization) and finally, Validation & Reporting. The following is a timeline of the group's development in chronological order from the feasibility studies to the deployment and documentation of the dashboard.

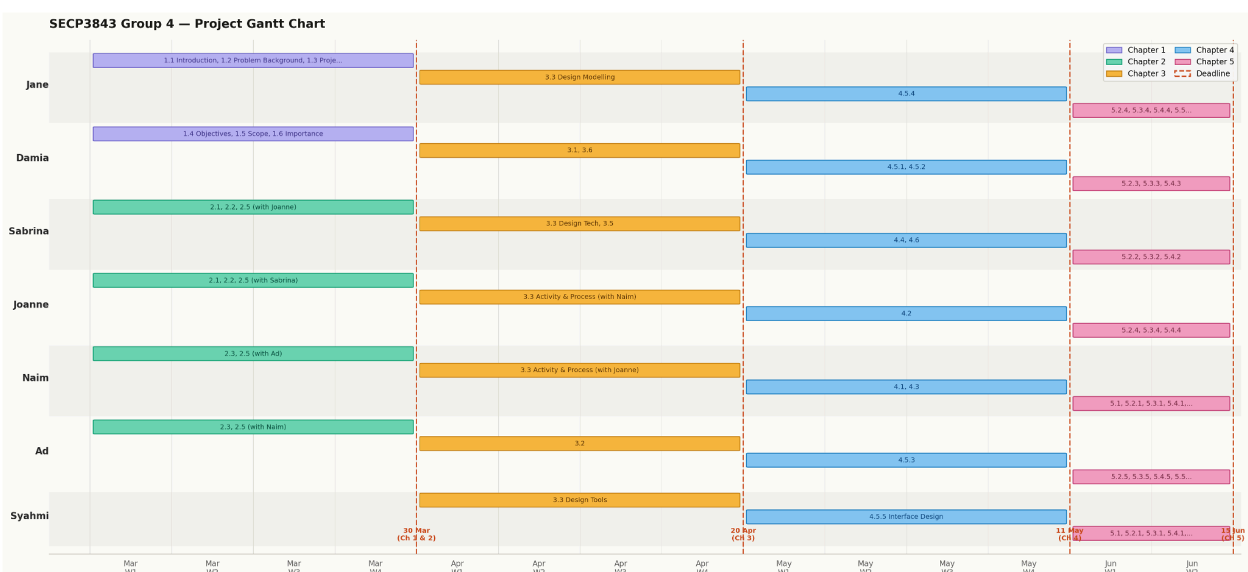


Figure 3.6.1 *Project Gantt Chart & Development Timeline*

3.7 Chapter Summary

This chapter discussed the methodology adopted for the development of the Inventory Risk Management (IRM) and Recoverable Assets (RA) system. The chosen methodology offers an organized and methodical way to guarantee that the project development process is executed successfully, efficiently and in accordance with the project's goals.

The chapter outlined the rationale behind using Medallion Architecture and the System Development Life Cycle (SDLC) methodology as the cornerstones of the suggested data engineering pipeline. The Bronze, Silver, and Gold levels of the Medallion Architecture facilitate the management of data flow from raw data ingestion through data transformation and cleansing to the creation of analytics-ready data for reporting and decision-making.

The many stages of the development process were also covered, including planning and requirement analysis, system design, development, testing, and deployment with assessment. Every stage was modified to meet the needs of the cloud-based data processing environment and the goals of inventory risk management.

The design modeling techniques, system specifications, and technologies utilized to facilitate the implementation of the suggested system were also covered in this chapter. Based on their responsibilities in data intake, storage, processing, transformation, and visualization, the features of Microsoft Azure services and Power BI were described.

All things considered, the approach described in this chapter provides a solid basis for the project's subsequent stage, which concentrates on system analysis, design, and implementation. The systematic approach ensures that the proposed solution can achieve the project's objectives while fostering future scalability, reliability and maintainability.

CHAPTER 4: ANALYSIS AND DESIGN

4.1 Introduction

This chapter is about the system analysis and design phases for the Recoverable Assets (RA) and Inventory Risk Management (IRM) system. The system analysis phase evaluates current operations to identify core business challenges and create the data collection plan required to build the data data pipeline. Based on these findings, the technical scope is outlined through detailed functional and non-functional requirements. Together, these elements form a blueprint that will be used to develop the multi layer azure pipeline and follow with power BI dashboard

4.2 System Analysis

This section examines the current PPG inventory management and what business problems to the problem. The analysis is conducted on the Current Process, the Datasets, Inventory Risk, and user requirements. The analysis outputs would determine the requirements and formulate a data engineering solution for inventory risk management.

4.2.1 Case Study

This project is based on a case study of PPG Coatings (Malaysia) Sdn. Bhd., a manufacturing company engaged in paints, coatings, and specialty materials. Due to the large-scale manufacturing and global supply chains of the industry, inventory management is crucial in ensuring the availability of materials and production activities are not slowed down (PPG Industries, 2024). PPG Coatings (Malaysia) Sdn. Bhd. Oversees the procurement, inventory operation, supply chain, and shared services in the Asia Pacific (AP) region (Malaysian Investment Development Authority [MIDA 2023]).

Information about the inventory such as stock levels, received suppliers' deliveries, consumption during production, expiry date etc, and customer demand data are stored in various operating systems. This disjointed system makes decision making and inventory visibility difficult. Findings in the project shows that there were inventory management issues in this project such as Excessive Inventory, Recoverable Assets (RA), Magna Carta (MC) Dead Stock, Expired Materials and Stockout risks that could slow production and affects customer orders.

In addition, there is a significant amount of inventory data, which is manually analysed and reported, which is likely to be time-consuming and error-prone. Improper inventory management and inability to account for dead inventory could result in profit loss, sub-optimal warehouse utilization, and delays in production (Gravier & Swartz, 2009).

To address these problems, a centralized inventory risk management system is proposed comprising the conjunction of many inventory databases into a data engineering pipeline with the use of business rules which display the inventory as RA, Magna Carta (MC) Dead Stock, Expired Materials, stockout risk, and respective customer sales orders on the dashboards to aid in decision making and inventory management.

4.2.2 System Requirements Gathering Techniques

The analysis of the project specification, the PPG supplied datasets, and the discussions during the industrial visit led to the derivation of the system requirements. The activities were used to understand the business objectives, inventory management problems, data requirements, and business analytical outputs of the proposed system.

An inventory management problem analysis was performed on the project specification to identify the problems with inventory management issues such as Recoverable Assets (RA), Magna Carta Dead Stock, Expired Materials, Stock out risks and their impact on production and customer orders faced by PPG.

Furthermore, the seven business questions for the proposed system were also identified in the specification. All 6 datasets supplied by PPG, namely materials.csv, inventory_snapshot.csv, production_orders.csv, purchase_orders.csv, suppliers.csv, and sales_orders.csv, were analysed to understand the data structures, attributes, and relationships between the datasets. Data requirements to support the identification of the data for the Bronze, Silver, Gold, and Curated layers were also developed as part of this analysis.

During the industrial visit to PPG, clarification discussions and Question & Answer were conducted with PPG representatives. In this direct interaction with the inventory business in PPG, data related to the business processes, inventory monitoring, business rules, and business analytical outputs information were gathered. The information collected was used to validate the system requirements and to ensure that the system solution was aligned with the requirements of the case study.

In general, the project specification analysis, data provided by PPG, and discussion during the industrial visit, enable the project team to identify the business objectives, informational requirements, and analytic requirements of the proposed system. The requirement gathering techniques employed in the industrial visit have aided the system developed to be in line with the Inventory management challenges and business expectations of the PPG case study.

4.3 System Requirements

This section explains both the functional and non-functional requirements of the system, which outline what the system needs to do and how it should perform

4.3.1 Functional Requirements

Table 4.3.1.1 outlines the functional requirements of the system, showing exactly what the software needs to do for the users.

Requirements Category	Description
Raw Data Ingestion (Bronze)	ingest 6 operational source CSV (materials.csv, inventory_snapshot.csv, purchase_orders.csv, production_orders.csv, suppliers.csv, and sales_orders.csv) into a secure cloud data lake while keeping their original raw formats.
Data Quality Guardrails (Silver)	Automatically validate data type,handle null values, and remove duplicate records
Recoverable Assets (RA)	Calculate 12 month based on the consumption rate and flag any inventory stock volume that exceeds this threshold as a Recoverable Asset (RA)
Magna Carta Dead Stock Flagging	Flag unexpired materials with more than equal 9 consecutive months of zero operational consumption
Magna Carta Expired Flagging	Compare active calendar timelines against raw lot parameters to flag it as Magna Carta Expired upon passing their designated lot expiry date
Financial Provisioning	Calculate a 100% financial write down provision for all flagged MC Dead and Expired Stock

Stockout Risk & Traceability	Flag materials running low relative supplier lead times, identify unfulfilled production orders, and trace the downstream impact to flag affected open customer sales orders and their associated clients
Star Schema Serving (Gold)	Structure processed attributes into a star schema, attaching all metrics around a central fact table.
Interactive Dashboards	The reporting interface must display final calculations by using a Power BI dashboard to solve the 7 specific business questions regarding inventory positions and risk metrics.

Table 4.3.1.1 *Functional Requirement*

4.3.2 Non Functional Requirements

Table 4.3.2.1 outlines the non-functional requirements that describe the overall quality and performance standards of the system

Requirements	Performance Standard
Data Integrity	Logic transformations used within cloud pipeline must maintain absolute data integrity, ensuring summaries perfectly match original source values
Query Performance	The dashboards Power BI must be optimized to render visual tiles and respond filtering queries within the acceptable operational limit
Scalability	The Azure infrastructure must capable to handling increasing volume of transactional snapshot rows over time without lowering the performance
Maintainability	Pipeline workflow within Microsoft Azure must follow structured design allowing engineering teams to easily update business logic formulas
Security and Access	Limit access to data lake, warehouse and reports dashboard to authenticated, authorized users

Table 4.3.2.1 Non Functional Requirement

4.4 Current System Analysis

Most manufacturing companies including the PPG industry case study use traditional and semi-manual inventory monitoring systems. These are typically Excel spreadsheets, simple ERP reporting applications and periodic manual reporting from various departments like procurement, warehouse and finance.

Typically, inventory data is not fully integrated in the current environment, and exists in isolated systems. For instance, production data is kept by the production team, sales data is handled by the sales team and inventory is controlled by the warehouse team. This results in a disjointed data model, with data not being updated in real time, and that is hard to integrate.

This has left managers with a lack of timely information to make decisions, as they have to manually integrate data from various sources to identify inventory risks, including excess stock, dead stock or stockout conditions. This increases the risks of human error and is inconsistent.

Another important limitation of the current system is the lack of predictive or automated inventory risk classification. For instance, items that are used for more than 12 months ago or items that have gone "out of stock" are not automatically marked as such and could only be discovered once losses have been incurred.

Otherwise, it is very manual and time-consuming to determine the effects of stockouts on production orders and customer sales orders. This lowers the efficiency of the operation and makes it hard for companies to react swiftly to disruptions in the supply chain.

So, there is a need for an enhanced automated data engineering pipeline. The proposed system with Microsoft Azure and Power BI will address these problems by consolidating all data sources within a single platform, adding automated business logic and offering real-time analytics dashboards for improved decision making.

4.5 System Design

Microsoft Azure services and Microsoft Power BI are used in cloud-based architecture of the proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) system. The Bronze, Silver and Gold layers of the Medallion Architecture guarantee effective data processing, scalability and data integrity. Monitoring inventory risks and making decision are made easier by the system's capacity to transform unprocessed inventory data into insightful business information.

4.5.1 System Architecture

Within its cloud-based architecture, the suggested Recoverable Assets (RA) and Inventory Risk Management (IRM) solution makes use of Microsoft Power BI. Through the Bronze, Silver and Gold layers of the Medallion Architecture design framework, the design makes data collecting, processing, storage and reporting functions easier.

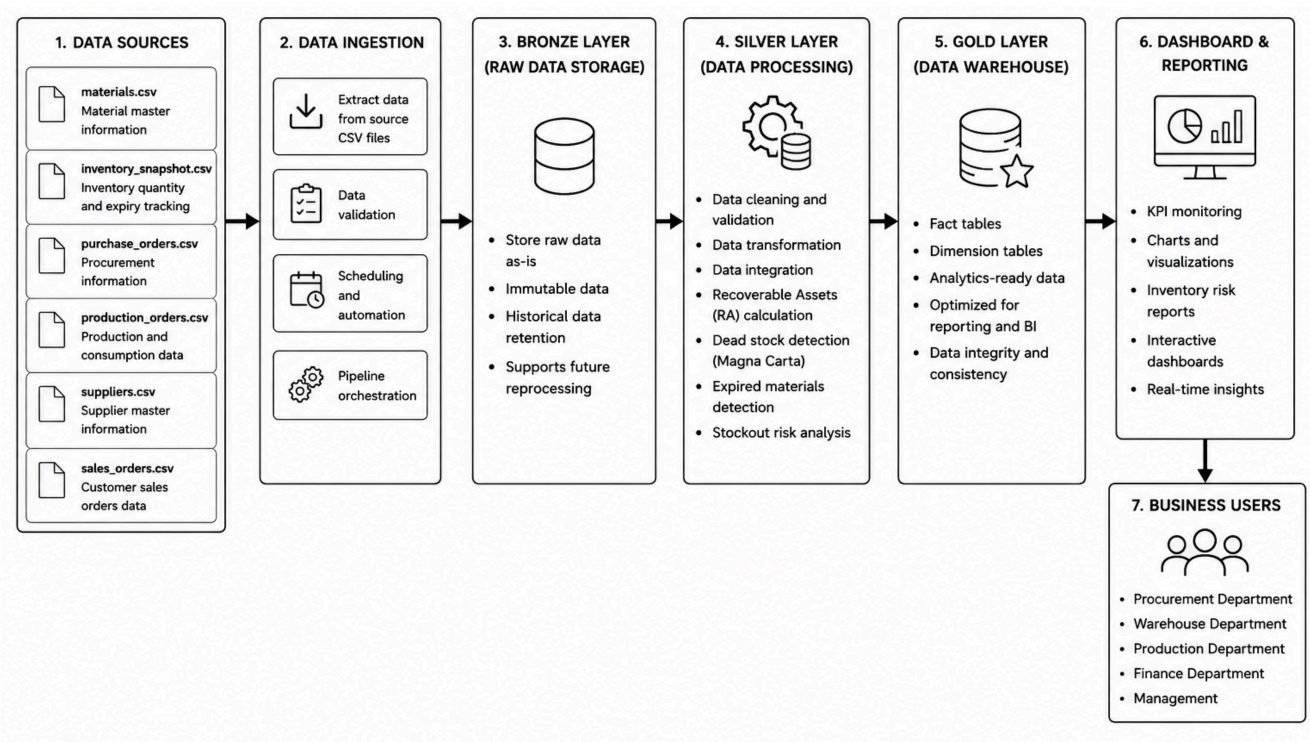


Figure 4.5.1.1 *System Architecture Diagram*

Figure 4.5.1.1 shows The Recoverable Assets (RA) and Inventory Risk Management (IRM) System's overall system architecture. The processed data is presented using dashboards and reports allowing different business users to make data-driven decisions. Each architectural components' individual functions are described in detail in the following subtopic.

System Flow

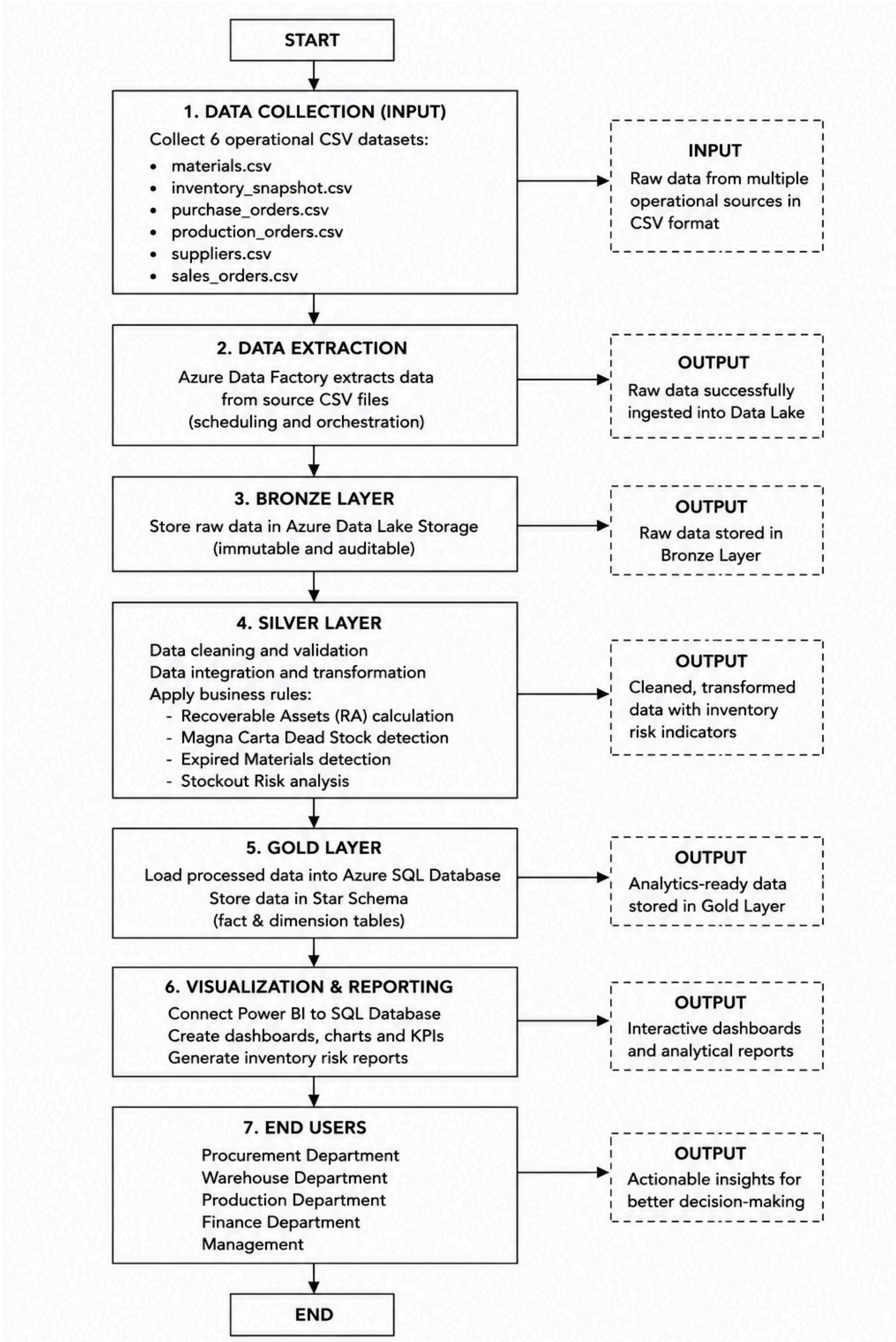


Figure 4.5.1.2 *System Flow Diagram*

The Recoverable Assets (RA) and Inventory Risk Management (IRM) System's system flow from data collection to end-user reporting is shown in Figure 4.5.1.2. First, operational datasets are collected which include CSV-formatted materials, inventory snapshots, purchase orders, manufacturing orders, suppliers and sales orders. The data extraction method is then used to retrieve the information and add it to the Azure Data Lake.

Before being cleaned, verified, integrated and processed in accordance with business rules in the Silver Layer, the raw data is first stored in the Bronze Layer. Important evaluations are carried out during this phase such as calculating recoverable assets, identifying dead stock, identifying expired goods and assessing stockout concerns. Following processing, the data is moved to the Gold Layer where it is arranged into tables in preparation for analysis and reporting. Lastly, Power BI dashboards and reports are used to display the data, providing business users including the departments of procurement, warehousing, production, finance and management with insightful information for making well informed decisions.

System Modules

Each functional module of the Recoverable Assets (RA) and Inventory Risk Management (IRM) System is in charge of a certain phase of the workflow for processing data. When taken as a whole, these modules make it easier to gather, store, process, analyze, report and make decisions. The primary modules and their corresponding roles inside the system are compiled in Table 4.5.1.3.

Modules	Functions
Data Ingestion Module	Extracts data from source CSV files and loads them into Azure Data Lake Storage
Data Storage Module	Stores raw inventory data in Bronze Layer
Data Processing Module	Cleans, transforms and validates inventory data
Inventory Risk Analysis Module	Identifies Recoverable Assets, Dead Stock, Expired Materials and Stockout Risks
Data Warehouse Module	Stores analytics-ready data in Star Schema format
Visualization	Displays KPIs, reports and dashboards through Power BI

Table 4.5.1.3 System Modules and Functions

System Inputs

The recommended Recoverable Assets (RA) and Inventory Risk Management (IRM) System makes use of six operational datasets provided by PPG Industries. These databases help with risk identification, inventory status evaluation and decision making.

The system receives data from six datasets:

- materials.csv
- inventory_snapshot.csv
- purchase_orders.csv
- production_orders.csv
- suppliers.csv
- sales_orders.csv

System Outputs

The system generates analytical reports and dashboards to support business decision-making and inventory risk management.

- The system generates:
- Recoverable Assets Report
- Magna Carta Dead Stock Report
- Expired Materials Report
- Stockout Risk Report
- Financial Provision Report
- Affected Customer Sales Orders Report
- Interactive Power BI Dashboards

4.5.2 Explanation of System Architecture Components

The Recoverable Assets (RA) and Inventory Risk Management (IRM) System are easier to create thanks to the system's design which consists of several interconnected components. In managing the complete data pipeline which comprises data collection, ingestion, storage, processing, transformation and visualization for decision-making, each component has a specific function. The Medallion Architecture concept which divides data handling into three primary layers, Gold, Silver and Bronze which is the foundation of the design.

The suggested system design is made up of a number of interconnected parts that make inventory risk management procedures easier. From data collection and storage to processing, analysis and reporting, every part of the data engineering pipeline has a distinct function. The duties related to each component are summarized here.

1. Data Collection (Input)

This component acts as the system's entry point. Six operational datasets, materials, inventory snapshots, purchase orders, production orders, supplier records, and sales orders are used to gather inventory-related data. These datasets include the crucial information required to evaluate inventory risk.

2. Data Extraction

Data extraction from the source CSV files is handled by Azure Data Factory (ADF). While automating the data entry process, it guarantees consistent and effective data collection. Additionally, ADF oversees the scheduling and orchestration of the entire data pipeline.

3. Bronze Layer

The Bronze Layer's raw data is kept in Azure Data Lake Storage (ADLS). The data is still in its original format. This layer serves as a central store for historical records and permits data audits and traceability.

4. Silver Layer

The data processing and transformation engine is Azure Databricks. Data is cleansed, verified, standardized and merged across several datasets in this layer. Additionally, inventory hazards are identified using business logic, including:

- Recoverable Assets (RA)
- Magna Carta Dead Stock
- Magna Carta Expired Materials
- Stockout Risks
- Financial Provision Calculations

Data accuracy, consistency and readiness for analytical processing are guaranteed by Silver Layer.

5. Gold Layer

Azure SQL Database serves as both a primary data warehouse and the Gold Layer. Fact and dimension tables are part of the Star Schema format which is used to organize processed data from Databricks. This method allows for efficient reporting and dashboard construction while also improving query efficiency.

6. Visualization and Reporting

Microsoft Power BI serves as the platform for business intelligence and visualization in the suggested system. After acquiring processed data from the Azure SQL Database, it presents information via interactive dashboards, charts, KPIs and reports. Real-time risk detection and inventory condition monitoring are made possible with Power BI.

7. End Users

Procurement officials, warehouse managers, production managers, finance analysts and senior management are among the system's end users. These clients use dashboard to make well informed decisions on operational risk management, stock management, buying plans and financial preparedness.

4.5.3 System Design

After touring PPG Coatings (Malaysia) Sdn. Bhd., our team developed an automated ELT (Extract, Load, Transform) pipeline using the Medallion Architecture to create complete reporting of a broken inventory system (Armbrust et al., 2021). This will allow them to preserve the non-destructive history of how they manage long-term risks for obsolete products through the ingestion of sis-core CSV datasets linked mainly by material_id into a bronze layer via Azure Data Factory (Gorelik, 2019; Gravier & Swartz, 2009).

Azure Databricks then cleanses and transforms this data in the Silver layer, applying specific PPG business logic to calculate Recoverable Assets and flag Magna Carta Dead Stock. Finally, the validated data is loaded into an Azure SQL Database as a dimensional star schema, Gold Layer (Kimball & Ross, 2013), allowing Microsoft Power BI to connect via Import Mode and deliver a high-performance, centralized dashboard for proactive logistics management (Microsoft, 2023). **Figure 4.5.3.1** illustrates this ELT orchestration flow.

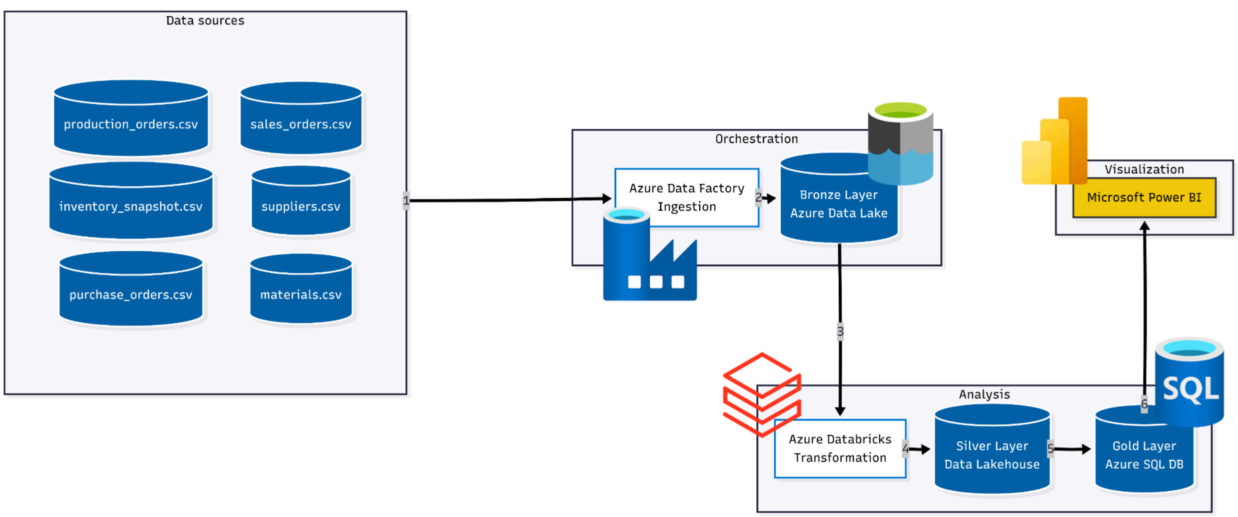


***Figure 4.5.3.1** ELT Orchestration Flow*

Data Engineering Process

The data engineering process is an automated, multi-hop pipeline that incrementally evolves data maturity by the Medallion Architecture (Armbrust et al., 2021). The first step is to load the static source files (the Bronze layer) into a data lake. By keeping this ingestion part of the pipeline separate from the transformation part, the environment will not turn into an unmanaged 'data swamp', with no metadata or governance (Gorelik, 2019) .

After being secured, data is transferred to the Silver layer for comprehensive data cleansing, type validates, and integration. It is at this point that the complicated PPG business logic is programmed – for example, the 12-month consumption limits are calculated and the Magna Carta thresholds are determined. Finally, the processed data is aggregated and organized in a dimensional star schema in Gold, for optimal performance of queries on the data.

Data Source

The pipeline uses synthetic operational information that is generated to mimic PPG's real-world supply chain. This data is available in six different Comma-Separated Values (CSV) files. In this schema, material_id is used as the primary relational link, it is the same key field and can be used in most of the data sets to track the material from supplier to customer from a comprehensive perspective:

- **materials.csv**: This is the base reference attributes table for materials, by category (Pigment, Resin, Solvent, Packaging) and is the primary dimension source.
- **inventory_snapshot.csv**: Records the stock levels and lot expiry dates every month. In this file, there is a significant amount of information that is essential for answering business question Q4 on Magna Carta Expired classifications.
- **purchase_orders.csv**: contains Inbound Transactions and late deliveries by supplier, which is needed for analyzing procurement efficiency.
- **suppliers.csv**: Stores the master information of the vendors such as geographic locations, reliability etc., which is closely associated with purchase orders data.
- **production_orders.csv**: This is a file that shows the production orders, and it is used to see if there are any unfulfilled production orders (Q6).
- **sales_orders.csv**: This cross-linking can be done with the inventory data using material_id so that the impact of the material stockout on "affected" customers sales can be traced (Q7).

Extraction, Transformation, and Loading (ETL)

To successfully execute the pipeline, it uses the coordination of certain Azure resources, coordinating the Extraction, Transformation, and Loading (ETL) steps to ensure they are executed correctly:

- **Extraction (Ingestion to Bronze)**: Azure Data Factory (ADF) is responsible for orchestrating the phase. The "Copy Data" activities are used to get the six CSV files that are the source data and store them as immutable in Azure Data Lake Storage (ADLS Gen2). This is the Bronze layer, which meets the need for a historical audit trail, with data being ingested without any loss of data and without any destruction (Gravier & Swartz, 2009).
- **Transformation (Refinement to Silver)**: Azure Databricks is used to perform the transformation phase (Refinement to Silver). Using PySpark and SQL notebooks that are connected to scalable compute clusters, data engineers clean the raw data. This stage deals with nulls, formatting dates to be consistent for lot expiry and the merging of the various CSV files with material_id. Computation of core business rules here, which creates definitive Recoverable Assets (RA) and Magna Carta Dead Stock flags.
- **Loading (Serving to Gold)**: This is the last step to validate and transform the data from Databricks and push to Azure SQL Database. The data is loaded into a star schema design that is pre-defined (Kimball & Ross, 2013). This schema has centralized Fact tables, that are surrounded by descriptive dimension tables like DimDate, DimMaterial and DimSupplier, namely FactInventoryStatus, FactConsumption, FactSales and FactProcurement.

Data Presentation and Visualization

Since the ELT pipeline runs on a scheduled batch-refresh cycle, as opposed to streaming real-time data, Import mode was chosen. This mode stores the data into memory, allowing for better visualization performance, and makes sure that the dashboard can quickly respond to logistics managers' unique business questions without incurring query latency (Microsoft, 2023).

4.5.4 Database Design

In the initial phase of database modeling, this project adopts a Non-SQL / Document-Oriented Data Modeling approach to structure the raw data ingested into the Bronze Layer of Azure Data Lake Storage. This non-relational (schemaless) method was selected so that the six source files belonging to the organization can be stored in their native formats instantly. This preserves historical data integrity through a non-destructive ingestion strategy.

In this Non-SQL model, data is not forced into rigid table rows and columns. Rather than utilizing a rigid relational framework, this architecture relies on independent document collections that flexibly establish logical connections via a shared material_id attribute. The structural outlines for each data collection are specifically tailored to their actual, real-world attributes.

The NoSQL database architecture for the Bronze Layer is divided into six distinct document collections, which house the foundational operational datasets for the inventory risk management system. To maximize schema adaptability and maintain the integrity of the source systems, each collection preserves data in its native format, a strategy that also enables highly scalable data ingestion into Azure Data Lake Storage. Furthermore, these foundational structures establish the primary attributes, data types, and business context necessary for all downstream activities, including data cleansing, transformation, and advanced analytics. The comprehensive schema specifications for these six collections are detailed in Tables 4.5.4.1 through 4.5.4.6.

a.materials.csv

Attribute	Data Type	Description
material_id	String	Unique identifier for the material
material_name	String	Name of the material
category	String	Material classification
unit_of_measure	String	Measurement unit
reorder_level	Number	Minimum stock to

		reorder
lead_time_delays	Number	Supplier delivery lead time in days
unit_cost	Number	Cost per unit of material

Table 4.5.4.1: Materials Collection Structure

b.inventory_snapshot.csv

Attribute	Data Type	Description
snapshot_id	String	Unique identifier for snapshot log
material_id	String	Reference key to the materials dataset
snapshot_date	Date	Date when the inventory snapshot was recorded
warehouse_location	Number	Physical storage location
quantity_on_hand	Number	Total physical stock available in the warehouse
expirey_date	Date	Material expiration date

Table 4.5.4.2:Inventory Snapshot Collection Structure

c.Production_orders.csv

Attribute	Data Type	Description
production_order_id	String	Unique identifier for the production run
material_id	String	Reference key to the materials dataset
production_date	Date	Date of the material consumption
finished_product	String	Name of the resulting output product
quantity_consumed	Number	Quantity of material used

Table 4.5.4.3:Production Orders Collection Structure

d.Sales_orders.csv

Attribute	Data Type	Description
sales_order_id	String	Unique identifier for the customer sales order
order_date	Date	Date when the customer places the order
material_id	String	Reference key to the materials dataset
customer_name	String	Name of the client or distributor
quantity_ordered	Number	Volume of product requested by customer

Table 4.5.4.4:Sales Orders Collection Structure

e.Purchase_orders.csv

Attribute	Data Type	Description
po_id	String	Unique identifier for the procurement order
material_id	String	Reference key to the materials dataset
supplier_id	String	Reference key to the suppliers dataset
po_date	Date	Date the purchase order was issued
expected_delivery_date	Date	Estimated arrival date promised by the vendor

Table 4.5.4.5:Purchase Orders Collection Structure

f.Suppliers

Attribute	Data Type	Description
supplier_id	String	Unique identifier for the vendor
supplier_name	String	Name of the supplier
country	String	Country of origin for the vendor
reliability_rating	Number	Performance rating score of the vendor

Table 4.5.4.6:Suppliers Collection Structure

4.5.5 Interface Design

This section describes the interface design of the Recoverable Assets (RA) and INventory Risk Management (IRM) dashboard. The dashboard acts as the final presentation layer of the system, converting processed data from the Gold Layer into meaningful visual information for procurement, warehouse, production, finance and management users. Microsoft Power BI Desktop was selected for the dashboard development because it integrates well with the Azure SQL Database and provides a wide range of interactive visualisation features.

The dashboard was developed through four main steps to ensure that it meets the projects business requirements:

- Step 1 - **Requirements Mapping:**
  Each business question from Q1 to Q7, was reviewed and matched with the most suitable visual representation. This ensured that every chart, KPI, and report element was designed to answer a specific business requirement and provide useful insight to users.
- Step 2 - **Wireframing:**
  Initial wireframes were created using Draw.io to plan the dashboard layout, KPI placement, and navigation structure. This helped establish a clear dashboard design before the development process began.
- Step 3 - **Data Connection and Modelling:**
  Power BI Desktop was connected to Azure SQL Database using Import mode. The relationships between the fact and dimension tables were verified in the Model View to ensure the star schema was implemented correctly. DAX measures were then created to calculate important KPIs such as Total MC Provision, RA Count, and Stockout Risk Count.
- Step 4 - **Visual Developmetn and Publishing:**
  The dashboard visuals were developed based on thep planned layout, and interactive features such as slicers, and drill-through functions were added to improve user experience. The dashboard was then tested against all seven business questions to ensure the results were accurate before being published to Power BI Service for end user access.

The dashboard is organised into four pages to avoid visual overload and to align each page with a specific user role, as summarised in Table 4.6.1.

Page	Title	Purpose and Business Questions Addressed
Page 1	Real-time Inventory Executive Summary	KPI cards for Total MC Provision, RA Count, Below-reorder Alerts, and At-risk Orders. Bar and donut charts by category. For senior management.
Page 2	Real-time Inventory Reorder Alert	Matrix of materials out of stock alignment + combination chart comparing available stock vs reorder levels. Q1. For procurement teams.
Page 3	Recoverable Assets Valuation	Materials classified as Recoverable Assets (Q2). Matrix of excess stock volume based on 12-month consumption and a treemap showing the monetary value (USD) of excesses. For management.
Page 4	Magna Carta (Dead Stock & Expired)	MC Dead Stock (Q3) and MC Expired (Q4). Side-by-side matrix of individual affected items and a stacked bar chart summarising total provision amount by provision type. For finance analysts.
Page 5	Financial Provision Breakdown	Total Magna Carta provision required (Q5). Waterfall chart breaking down provision by material category and a detailed matrix listing specific dead and expired materials. For finance and stakeholders.
Page 6	Stockout Risk and Customer Impact	Unfilled production orders (Q6) and affected open sales orders and customers (Q7). Summary matrix of at-risk orders and a detailed table with conditional formatting to highlight urgent

		delivery dates. For the logistics and customer service team.

Table 4.6.1: *Dashboard page structure and alignment with the seven business questions* .

The choice of chart type follows the principle that the visual representation must match the nature of the underlying data and the question being asked. Table 4.6.2 summarises the mapping between business questions and chart types.

ID	Business Question	Chart Type	Justification
Q1	Materials below reorder level	Matrix + combination chart	Matrix shows exact stock misalignment per material; combination chart visually compares available stock against the reorder threshold.
Q2	Materials classified as Recoverable Assets	Matrix + treemap	Matrix details excess volume based on 12-month consumption; treemap surfaces high-value materials with trapped capital that can be liquidated or reallocated.
Q3	Materials qualifying as MC Dead Stock	Side-by-side matrix + stacked bar chart	Matrix gives granular item-level detail; stacked bar chart provides a weighted summary of provision by type for quick financial assessment.
Q4	Materials qualifying as MC Expired	Side-by-side matrix + stacked bar chart	Combined with Q3 on the same page; matrix identifies individual expired items and stacked bar chart shows the financial breakdown by provision type.

Q5	Total Magna Carta provision required	Waterfall chart + detailed matrix	Waterfall chart communicates the step-by-step build-up of provision by material category; matrix enables item-wise tracking of actual inventory losses.
Q6	Unfulfilled production orders	Summary matrix + detailed table with conditional formatting	Matrix consolidates at-risk order quantities; conditional formatting automatically highlights overdue or near-due delivery dates in red for immediate action.
Q7	Open sales orders and customers affected by shortage	Detailed table with conditional formatting	Detailed table traces impacted sales orders and customers downstream; urgency indicator flags the most critical deliveries to support proactive client management.

Table 4.6.2: Mapping of business questions to chart types and justification.

4.6 Chapter Summary

The system analysis and design of the proposed Recoverable Assets (RA) and Inventory Risk Management (IRM) system was discussed in this chapter. The first step in the analysis was to look at the existing inventory management practices and pinpoint a number of challenges, such as the lack of integration between data sources, the manual monitoring processes, limited real-time visibility and the inefficiency in identifying inventory risks. With the basis of these findings, the functional and non-functional requirements of the proposed system were determined.

In addition, the chapter presented the overall system architecture and design of the solution. The proposed system will be based on the Medallion Architecture, which includes the Bronze, Silver and Gold layers for data ingestion, transformation, storage and analytics. The following

Microsoft Azure Services were chosen to create an integrated data engineering and business intelligence platform.

In addition, the chapter explained the system flow, system database design, data processing modules and dashboard interface design. The proposed solution is capable of identifying Recoverable Assets, Magna Carta Dead Stock, Magna Carta Expired Materials, stockout risks, financial provision and customer order impacts via automatic data processing and reporting. The dashboard design was tailored to the identified business requirements, ensuring that users can easily access and track inventory conditions and make informed decisions.

Overall, the proposed system design provides a structured, scalable, and efficient approach to inventory risk management. The design serves as the foundation for the implementation phase, which will be discussed in the following chapter.

CHAPTER 5 : IMPLEMENTATION AND TESTING

5.1 Introduction

This Chapter discusses the implementation and testing of the Pipeline cloud based inventory analysis system for the PPG industry. The implementation phase is focused on the use of Microsoft Azure service based on the Medallion Architecture framework to create the system. The system ingest multiple files of dataset, process it through multiple layers and create insightful information by interactive dashboard. Furthermore, to make sure the system reaches the expectation of the company needs, the tests which is Black Box are being conducted to evaluate the functionality of the system. The result of this test will verify either the system is running as expected or not.

5.2 Coding of System Main Functions

This part outlines the details of the code for each main function and explains how every single part of the pipeline works together.

5.2.1 Bronze Layer - Azure Setup & Ingestion

The bronze layer is created to store the raw data without changing it, making it easy for the silver layer to fetch the data and perform data preprocessing tasks such as cleaning and filtering.

Azure Data Factory

Figure 5.2.1.1 shows the Azure Data Factory pipeline named pl_bronze_ingest. The pipeline contains six Copy Data activities, one for each CSV file (materials, inventory, purchase_orders, production_orders, suppliers, and sales_orders). Each activity copies one CSV file from the source location into the Bronze container in Azure Data Lake Storage. This pipeline is the main entry point of the system and is used to load all raw data into the cloud in one run.

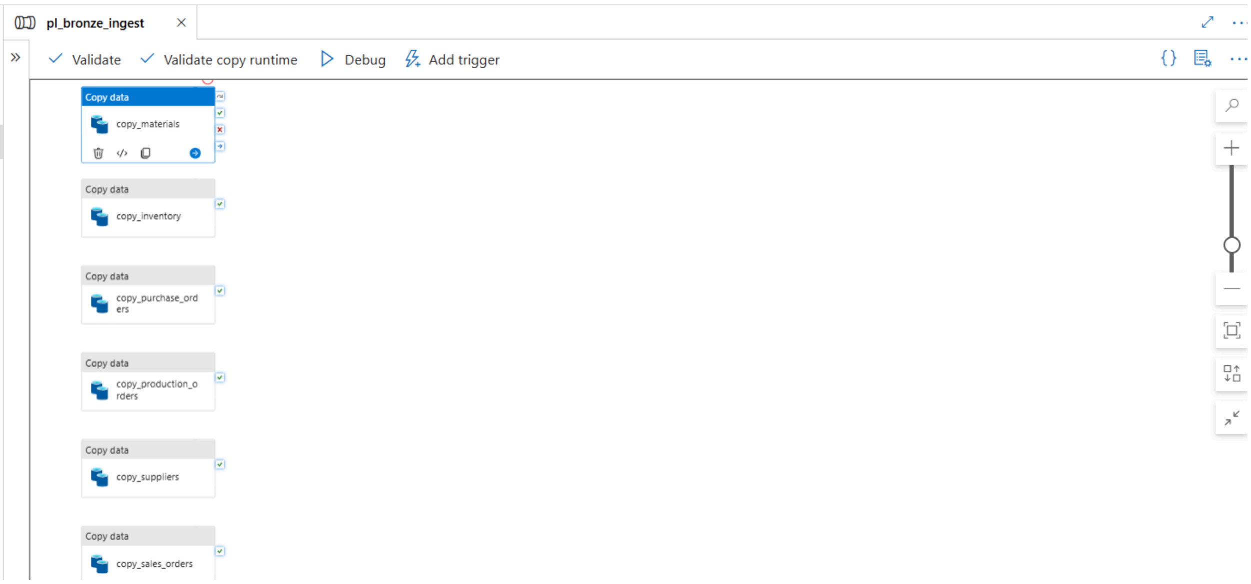


Figure 5.2.1.1 Azure Data Factory pipeline

Figure 5.2.1.1 shows the Azure Data Factory pipeline named pl_bronze_ingest. The pipeline contains six Copy Data activities, one for each CSV file (materials, inventory, purchase_orders, production_orders, suppliers, and sales_orders). Each activity copies one CSV file from the source location into the Bronze container in Azure Data Lake Storage. This pipeline is the main entry point of the system and is used to load all raw data into the cloud in one run.

Audit Query 1 - materials

```sql
SELECT
    COUNT(*)                                                          AS total_rows,
    COUNT(DISTINCT material_id)                                       AS distinct_material_ids,
    SUM(CASE WHEN material_id IS NULL THEN 1 ELSE 0 END)              AS null_material_ids,
    SUM(CASE WHEN material_name IS NULL THEN 1 ELSE 0 END)            AS null_material_names,
    SUM(CASE WHEN unit_cost IS NULL THEN 1 ELSE 0 END)                AS null_unit_costs,
    SUM(CASE WHEN reorder_level IS NULL THEN 1 ELSE 0 END)            AS null_reorder_levels,
    COUNT(DISTINCT category)                                          AS distinct_categories
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/materials/materials.csv',
    FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
) AS r;
```

Figure 5.2.1.2 Coding for Audit Query 1

Figure 5.2.1.2 shows the SQL audit query for the materials.csv file. To read the file directly from the bronze container, the query OPENROWSET is used. The code counts the total row, checks for null values for the important column and counts how many categories it has. The result shows that 20 materials were loaded with no missing values and a total of 4 categories.

Audit Query 2 - inventory_snapshot

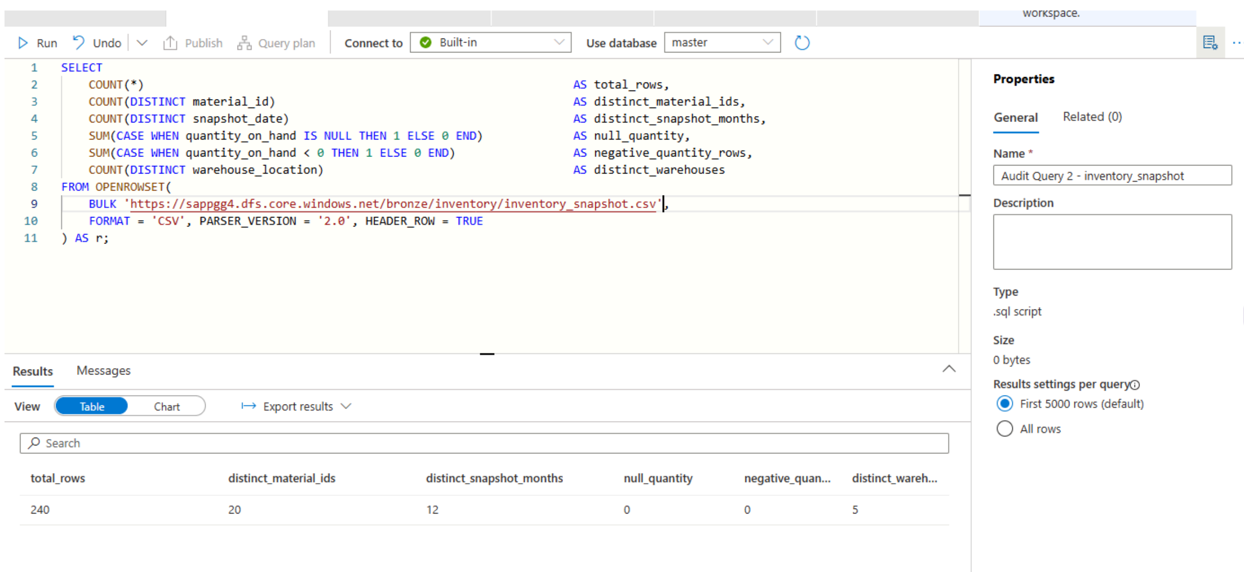


Figure 5.2.1.3 Coding for Audit Query 2

Figure 5.2.1.3 shows the audit code for the inventory_snapshot.csv file. The query checks the total number of rows, number of snapshot for each month, and the null value. The result shows that the total rows is 240 across each month with no null value. This confirms that the monthly inventory data was ingested successfully and ready to be used for the next layer

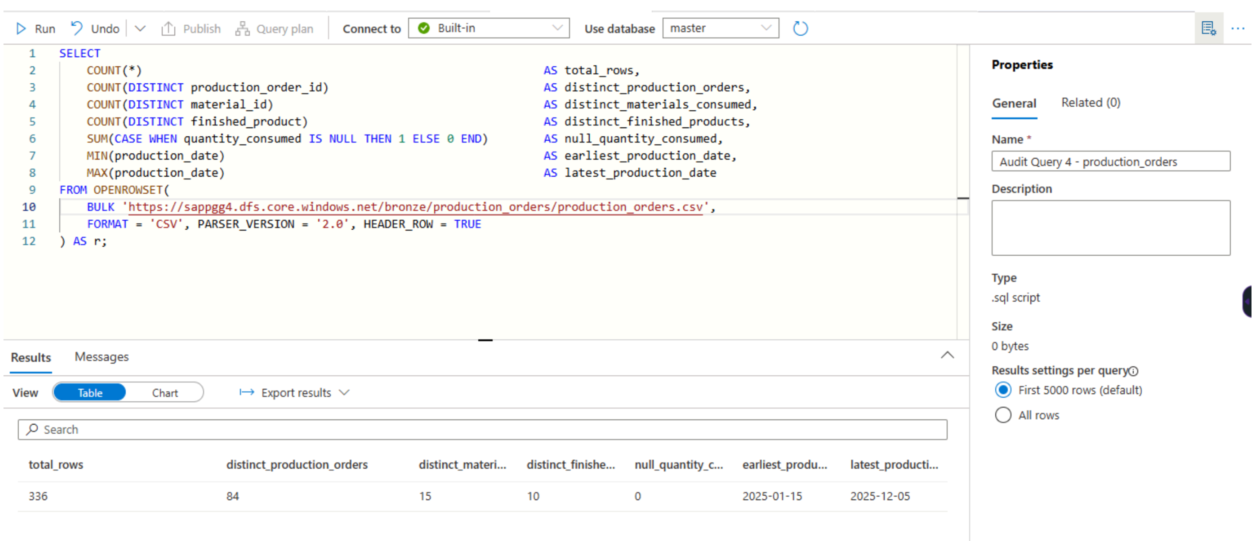
Audit Query 3 - purchase_orders

```sql
SELECT
    COUNT(*)                                                                    AS total_rows,
    COUNT(DISTINCT po_id)                                                       AS distinct_po_ids,
    SUM(CASE WHEN actual_delivery_date = '' THEN 1 ELSE 0 END)                  AS open_undelivered_pos,
    SUM(CASE WHEN CAST(quantity_received AS INT) > CAST(quantity_ordered AS INT)
            THEN 1 ELSE 0 END)                                                  AS over_delivered_rows,
    SUM(CASE WHEN CAST(quantity_received AS INT) < CAST(quantity_ordered AS INT)
             AND actual_delivery_date != ''
            THEN 1 ELSE 0 END)                                                  AS partial_delivery_rows
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/purchase_order/purchase_orders.csv',
    FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
) AS r;
```

Figure 5.2.1.4 Coding for Audit Query 3

Figure 5.2.1.4 shows the audit query for the purchase_orders.csv file. This code counts the total rows, the unique purchase order ID and spot the open undelivered purchase order, over delivered rows, and partial deliveries. The result shows that 40 purchases were ingested with one open undelivered PO. This code confirm that the data is complete and the special delivery situations can be detected

Audit Query 4 - production_orders




```sql
SELECT
    COUNT(*)                                                        AS total_rows,
    COUNT(DISTINCT production_order_id)                             AS distinct_production_orders,
    COUNT(DISTINCT material_id)                                     AS distinct_materials_consumed,
    COUNT(DISTINCT finished_product)                                AS distinct_finished_products,
    SUM(CASE WHEN quantity_consumed IS NULL THEN 1 ELSE 0 END)      AS null_quantity_consumed,
    MIN(production_date)                                            AS earliest_production_date,
    MAX(production_date)                                            AS latest_production_date
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/production_orders/production_orders.csv',
    FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
) AS r;
```

total_rows	distinct_production_orders	distinct_materi...	distinct_finishe...	null_quantity_c...	earliest_produ...	latest_producti...
336	84	15	10	0	2025-01-15	2025-12-05

Figure 5.2.1.5 Coding for Audit Query 4

Figure 5.2.1.5 shows the audit query for the production_orders.csv file. The code counts the total rows, the distinct production order ID, distinct materials consumed, district finished product, and the production dates. The result shows that there are 336 rows with 10 finished products, and no null values in the quantity consumed column. This will prove that the production data is ready to use to make analysis

Audit Query 5 - suppliers

```
SELECT
    COUNT(*)                                                     AS total_rows,
    COUNT(DISTINCT supplier_id)                                  AS distinct_supplier_ids,
    SUM(CASE WHEN supplier_id IS NULL THEN 1 ELSE 0 END)         AS null_supplier_ids,
    COUNT(DISTINCT country)                                      AS distinct_countries,
    COUNT(DISTINCT reliability_rating)                           AS distinct_ratings
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/suppliers/suppliers.csv',
    FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
) AS r;
```

Figure 5.2.1.6 Coding for Audit Query 5

Figure 5.2.1.6 illustrates the audit query applied to supplier.csv, covering total rows, distinct IDs, countries, and reliability ratings. The results reveal 10 suppliers from 7 countries with 3 distinct reliability ratings and no null supplier IDs, establishing that the reference data is complete and suitable for procurement analysis via purchase order linkage.

Audit Query 6 - sales_orders

```
SELECT
    COUNT(*)                                                     AS total_rows,
    COUNT(DISTINCT sales_order_id)                               AS distinct_order_ids,
    SUM(CASE WHEN status = 'Open' THEN 1 ELSE 0 END)             AS open_orders,
    SUM(CASE WHEN status = 'Delivered' THEN 1 ELSE 0 END)        AS delivered_orders,
    COUNT(DISTINCT customer_name)                                AS distinct_customers,
    COUNT(DISTINCT finished_product)                             AS distinct_products,
    MIN(required_delivery_date)                                  AS earliest_delivery,
    MAX(required_delivery_date)                                  AS latest_delivery
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/sales_order/sales_orders.csv',
    FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
) AS r;
```

Figure 5.2.1.7 Coding for Audit Query 6

The audit query shown in Figure 5.2.1.7 examines sales_orders.csv by counting total rows, unique sales order IDs, Open and Delivered order counts, and distinct customers and finished products. With 48 orders confirmed , 10 Open and 38 Delivered , this dataset directly supports Q7, which investigates the impact of stockout risks on outstanding customer sales orders.

5.2.2 Silver Layer - Data Cleaning

The Silver Layer was added to enhance the quality and consistency of the raw data that is stored in the Bronze Layer. It is mainly used to clean, standardize and transform the data before it is used for further analysis in the Gold Layer.

Silver 0 - Setup

Figure 5.2.2.1 shows a script called Silver 0 - Setup that is developed to setup the Silver Layer environment. This script was used to create the ppg_silver database and to setup the external data source and file format needed to store the transformed datasets in Parquet format.

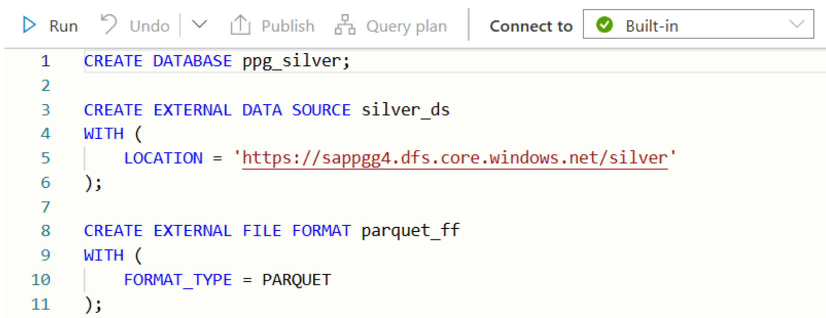


Figure 5.2.2.1: Silver Layer Setup Script Execution

After setting up, there were six CETAS (Create External Table As Select) scripts. The six datasets collected by the Bronze Layer were processed into a single dataset using the following procedures. Each script reads data from a different file.

Silver Materials CETAS

Figure 5.2.2.2 shows the first transformation script that was created for the Materials dataset. During this process, unnecessary spaces in text fields such as material identifiers, material names, and categories have been removed using the TRIM() function. Otherwise, CAST() functions have converted the numeric attributes including unit cost, reorder level, and lead time into appropriate data types. Duplicate records were also eliminated by the DISTINCT keyword and records containing missing material were eliminated by the WHERE clause.

Run Undo Publish Query plan Connect to Built-in Use database ppg_silver

```sql
CREATE EXTERNAL TABLE silver_materials
WITH (
    LOCATION = 'materials/',
    DATA_SOURCE = silver_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT DISTINCT
    TRIM(material_id) AS material_id,
    TRIM(material_name) AS material_name,
    TRIM(category) AS category,
    CAST(unit_cost AS FLOAT) AS unit_cost,
    CAST(reorder_level AS INT) AS reorder_level,
    CAST(lead_time_days AS INT) AS lead_time_days
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/materials/materials.csv',
    FORMAT = 'CSV',
    PARSER_VERSION = '2.0',
    HEADER_ROW = TRUE
) AS r
WHERE material_id IS NOT NULL
AND material_name IS NOT NULL;
```

Figure 5.2.2.2: Materials Transformation Script

Silver Inventory CETAS

Figure 5.2.2.3 shows the second script processed in the Inventory dataset. The date of the snapshot was transformed into the DATE format and the quantities of inventory were converted into integer values. In addition, records containing missing material identifiers or negative inventory quantities were filtered out to ensure that only valid inventory records were retained.

Run Undo Publish Query plan Connect to Built-in Use database

```sql
CREATE EXTERNAL TABLE silver_inventory
WITH (
    LOCATION = 'inventory/',
    DATA_SOURCE = silver_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT DISTINCT
    CAST(snapshot_date AS DATE) AS snapshot_date,
    TRIM(material_id) AS material_id,
    CAST(quantity_on_hand AS INT) AS quantity_on_hand,
    TRIM(warehouse_location) AS warehouse_location
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/inventory/inventory_snapshot.csv',
    FORMAT='CSV',
    PARSER_VERSION='2.0',
    HEADER_ROW=TRUE
) AS r
WHERE material_id IS NOT NULL
AND quantity_on_hand >= 0;
```

Figure 5.2.2.3: Inventory Transformation Script

Silver Purchase Orders CETAS

Figure 5.2.2.4 shows that the Purchase Orders dataset was then normalized by cleaning up the text fields and converting them to numbers. The script also shows that it converts quantity and date attributes to the correct data types. This makes sure that in the later stages of the analysis can be used to better analyze information.

Run | Undo | Publish | Query plan | Connect to Built-in | Use database ppg_silver

```sql
CREATE EXTERNAL TABLE silver_purchase_orders
WITH (
    LOCATION = 'purchase_orders/',
    DATA_SOURCE = silver_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT
    TRIM(po_id) AS po_id,
    TRIM(material_id) AS material_id,
    TRIM(supplier_id) AS supplier_id,
    CAST(quantity_ordered AS INT) AS quantity_ordered,
    CAST(quantity_received AS INT) AS quantity_received,
    CAST(po_date AS DATE) AS po_date,
    CAST(expected_delivery_date AS DATE) AS expected_delivery_date
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/purchase_order/purchase_orders.csv',
    FORMAT='CSV',
    PARSER_VERSION='2.0',
    HEADER_ROW=TRUE
) AS r;
```

Figure 5.2.2.4: Purchase Orders Transformation Script

Silver Production Orders CETAS

Figure 5.2.2.5 is the Production Orders dataset, unnecessary spaces were removed from text fields and the quantity consumed and production date attributes were transformed to appropriate formats. This helps improve data consistency and supports future analytical processing.

```sql
CREATE EXTERNAL TABLE silver_production_orders
WITH (
    LOCATION = 'production_orders/',
    DATA_SOURCE = silver_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT
    TRIM(production_order_id) AS production_order_id,
    TRIM(finished_product) AS finished_product,
    TRIM(material_id) AS material_id,
    CAST(quantity_consumed AS INT) AS quantity_consumed,
    CAST(production_date AS DATE) AS production_date
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/production_orders/production_orders.csv',
    FORMAT='CSV',
    PARSER_VERSION='2.0',
    HEADER_ROW=TRUE
) AS r;
```

Figure 5.2.2.5: Production Orders Transformation Script

Silver Suppliers CETAS

Figure 5.2.2.6 shows the Suppliers dataset was cleaned by eliminating duplicate records and standardizing supplier-related information. Records with missing supplier identifiers were excluded to maintain data integrity.

```sql
CREATE EXTERNAL TABLE silver_suppliers
WITH (
    LOCATION = 'suppliers/',
    DATA_SOURCE = silver_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT DISTINCT
    TRIM(supplier_id) AS supplier_id,
    TRIM(supplier_name) AS supplier_name,
    TRIM(country) AS country,
    TRIM(reliability_rating) AS reliability_rating
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/suppliers/suppliers.csv',
    FORMAT='CSV',
    PARSER_VERSION='2.0',
    HEADER_ROW=TRUE
) AS r
WHERE supplier_id IS NOT NULL;
```

Figure 5.2.2.6: Suppliers Dataset Transformation Script

Silver Sales Orders CETAS

Lastly Figure 5.2.2.7 shows the Sales Orders dataset was transformed by standardizing text fields and converting quantity and date attributes into the correct formats. This guarantees that customer order information remains consistent and available for analysis.

Run Undo Publish Query plan Connect to Built-in Use database

```sql
CREATE EXTERNAL TABLE silver_sales_orders
WITH (
    LOCATION = 'sales_orders/',
    DATA_SOURCE = silver_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT
    TRIM(sales_order_id) AS sales_order_id,
    TRIM(customer_name) AS customer_name,
    TRIM(finished_product) AS finished_product,
    CAST(quantity_ordered AS INT) AS quantity_ordered,
    CAST(order_date AS DATE) AS order_date,
    CAST(required_delivery_date AS DATE) AS required_delivery_date,
    TRIM(status) AS status
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/sales_order/sales_orders.csv',
    FORMAT='CSV',
    PARSER_VERSION='2.0',
    HEADER_ROW=TRUE
) AS r;
```

Figure 5.2.2.7: Sales Orders Dataset Transformation Script

Once all the transformation activities were finised, the cleaned datasets were exported to the Silver container based on CETAS approach. The datasets were stored in Parquet format because it provides better storage efficiency and improved query performance over traditional CSV files. These cleaned datasets are the basis of the Gold Layer, which includes business metrics, KIPs and dashboard visualizations are generated.

5.2.3 Gold Layer - RA/MC Business Logic

This section explains how the PPG Inventory Pipeline project uses the Gold Layer business logic to facilitate financial provisioning and assess inventory risk. The Gold Layer uses a set of business rules and integrates information from the Silver Layer to determine which commodities need inventory management attention. These industry standards provide the foundation for the answers to business questions Q1 to Q5.

Twelve-Month Consumption Calculation

This consumption first calculates the expected inventory utilization over a 12-month period in order to determine Recoverable Assets (RA). This computation is essential because materials that exceed their annual consumption requirements could indicate excess inventory levels and potential stock management inefficiencies. By looking at past consumption data, the script determines the required consumption level for RA classification.

```sql
CREATE OR ALTER VIEW dbo.gold_consumption_12m
AS

SELECT
    material_id,
    SUM(quantity_consumed) AS consumption_12m
FROM dbo.silver_production_orders
WHERE production_date >= DATEADD(day,-365,'2025-12-31')
GROUP BY material_id;
```

Figure 5.2.3.1: Gold Consumption 12M Script

The Gold Layer script used to determine the annual consumption value is shown in Figure 5.2.3.1. By gathering material consumption statistics, the script calculates the expected annual usage quantity for each material. This calculated amount is used as a benchmark to determine whether the present stock level exceeds typical consumption needs. This technique eliminates the need for manual inspections by allowing the system to identify goods that might be recovered assets on its own.

Latest Inventory Aggregation

The system performs inventory consolidation to obtain the most recent inventory status for each item after determining consumption requirements. Before implementing RA and MC business laws, this process combines inventory data and consolidates the current stock level to provide a new view on material accessibility.

```
CREATE OR ALTER VIEW dbo.gold_latest_inventory
AS

SELECT
    material_id,
    SUM(quantity_on_hand) AS current_stock,
    MAX(snapshot_date) AS as_of_date
FROM dbo.gold_inventory_extended
WHERE snapshot_date <= '2025-12-31'
GROUP BY material_id;
```

Figure 5.2.3.2: Gold Latest Consumption Script

Figure 5.2.3.2 displays the script used to categorize Recoverable Assets. Using the twelve-month consumption norm, the script looks at the most recent inventory levels. Recoverable assets (RA) are materials with inventory levels greater than the required consumption quantity. The business may quickly discover excess inventory and improve inventory efficiency thanks to this automatic classification.

Inventory Risk Classification Flags

The coding used to classify Recoverable Assets is displayed in Figure 5.2.3.2. Using the twelve-month consumption norm, the script looks at the most recent inventory levels. Recoverable assets (RA) are materials whose inventory levels exceed the required consumption quantity. The company can quickly discover excess stock and improve inventory usage thanks to this automated categorization.

Q1 - Below Reorder Flag

Materials with insufficient inventory levels relative to their required reorder point are indicated by the reorder flag below. The hazards associated with inventory shortfalls that may affect production operations are identified by this rule.

```sql
-- Q1
CASE WHEN i.current_stock < m.reorder_level THEN 1 ELSE 0 END AS below_reorder,

```

Figure 5.2.3.3: Gold Q1 Fact Inventory Status Script

The code that applies the reasoning for reorder-level verification is shown in Figure 5.2.3.3. When supplies fall below the required amount, the system tracks inventory levels and notifies the designated reorder point. By identifying items that may require refilling, this strategy enables proactive inventory oversight.

Q2 - Recoverable Asset Flag

Finding products where the current inventory exceeds the consumption requirements for the upcoming year is the primary goal of the Recoverable Asset (RA) classification. These materials are excess inventory with potential value that has to be kept an eye on to avoid going out of style or being underutilized.

The following logic was implemented:

```sql
-- Q2
CASE WHEN i.current_stock > ISNULL(c.consumption_12m,0) THEN 1 ELSE 0 END AS is_recoverable_asset,

```

Figure 5.2.3.4: Gold Q2 Fact Inventory Status Script

Figure 5.2.3.4 illustrates the implementation of the Recoverable Asset classification system. The script assesses every material by contrasting the inventory amount with the computed twelve-month usage value. When the stock quantity exceeds the necessary usage limit, the material is categorized as a Recoverable Asset. This automated identification enhances the awareness of excess inventory situations.

Q3 - Magna Carta(MC) Dead Stock Flag

Items that might no longer be pertinent for operational tasks after being unused for an extended duration are recognized utilizing the Magna Carta Dead Stock standard. Identifying dead stock is crucial because surplus inventory can raise storage costs and lead to monetary losses.

```sql

-- Q3
CASE
    WHEN (l.last_consumed_date IS NULL OR DATEDIFF(MONTH, l.last_consumed_date, '2025-12-31') >= 9)
         AND e.lot_expiry_date IS NULL
    THEN 1 ELSE 0
END AS is_mc_dead_stock,
```

Figure 5.2.3.5: Gold Q3 Fact Inventory Status Script

Figure 5.2.3.5 illustrates the code employed to detect Magna Carta Dead Stock components. The logic analyzes inventory shifts and usage records to identify products that quality as dead stock. The resources specified by this regulation are intended for additional assessment and potential financing.

Q4 - Magna Carta(MC) Expired Flag

Expired material detection identifies inventory items that have surpassed their expiration date and are no longer usable. Since expired items represent a direct loss in inventory and must be appropriately managed financially, this calculation is essential.

```sql

-- Q4
CASE
    WHEN e.lot_expiry_date IS NOT NULL AND e.lot_expiry_date < '2025-12-31'
    THEN 1 ELSE 0
END AS is_mc_expired,
```

Figure 5.2.3.6: Gold Q4 Fact Inventory Status Script

Figure 5.2.3.6 displays the script used to identify expired objects. The system compares the recorded expiration date of each inventory item with the current date. When the expiration condition is met, the substance is automatically classified as expired. This reduces the need for human verification and improves inventory monitoring accuracy.

Q5 - Magna Carta(MC) Provision Calculation

The anticipated financial effect of outdated inventory is established by the provision calculation. The algorithm calculates the required provision amount to capture the possible financial loss for items designated as Magna Carta Dead Stock or Expired Materials.

```
    -- Q5
    CASE
    WHEN
    (
        (
            l.last_consumed_date IS NULL
            OR
            DATEDIFF(MONTH,l.last_consumed_date,'2025-12-31') >= 9
        )
        AND e.lot_expiry_date IS NULL
    )
    OR
    (
        e.lot_expiry_date IS NOT NULL
        AND e.lot_expiry_date < '2025-12-31'
    )
    THEN i.current_stock * m.unit_cost
    ELSE 0
END AS mc_provision_amount,
```

Figure 5.2.3.7: Gold Q5 Fact Inventory Status Script

Figure 5.2.3.7 illustrates how the Gold Layer calculates provisions. The script calculates the financial provision amount based on the value of the affected inventory items. This calculation helps management teams understand the potential repercussions of out-of-date materials and provides them with financial transparency.

5.2.4 Gold Layer Extension and Query 6 & 7

D1 Data Extension – Adding lot_expiry_date

Not all attributes used by Magna Carta Expired (Q4) analysis are available in the original inventory data set.In order to solve this problem, a "Gold Layer" view named gold_inventory_extended has been created.The goal is to add the expiration date information to the inventory.The expiration date is calculated according to a set of business rules. The expiration date of materials MAT001 and MAT004 is set to 2025-06-01 to mark invalid items, while materials MAT016 to MAT020 are set to NULL, indicating that these materials expiration status was not applicable.The expiry date for all remaining materials is 2026-12-31, which is interpreted that had not yet expired.

```
CREATE OR ALTER VIEW dbo.gold_inventory_extended
AS
SELECT
    snapshot_date,
    material_id,
    quantity_on_hand,
    warehouse_location,
    CASE
        WHEN material_id IN ('MAT001','MAT005')
            THEN CAST('2025-06-01' AS DATE)

        WHEN material_id IN
        (
            'MAT016',
            'MAT017',
            'MAT018',
            'MAT019',
            'MAT020'
        )
            THEN NULL

        ELSE CAST('2026-12-31' AS DATE)

    END AS lot_expiry_date

FROM ppg_silver.dbo.silver_inventory;
```

Figure 5.2.4.1: Gold Inventory Extension (D1) Script

The SQL script that creates the "lot_expiry_date" attribute is shown in Figure 5.2.4.1. This script uses conditional business rules based on material categories using a CASE statement to set the expiration date, future expiration date and NULL.

Probability Distribution Explanation

The expiry date assignment was designed to simulate realistic inventory management scenarios. Packaging and non-expiry-controlled materials were assigned NULL values because expiry tracking is not applicable. Two materials were intentionally assigned expired dates to support Magna Carta Expired (Q4) testing, while the remaining materials were assigned future expiry dates to represent valid inventory.The distribution of expiry categories used in the D1 data extension is summarized in Table 5.2.4.1.

Table 5.2.4.1: Distribution of Generated Expiry Date Categories

Expiry Category	Materials	Count	Percentage
Expired	MAT001, MAT005	2	10%
No Expiry Tracking	MAT016-MAT020	5	25%
Future Expiry	Remaining Materials	13	65%

Q6-Unfulfilled production query logic

The objective of Q6 was to determine whether any production orders could not be fulfilled due to insufficient inventory.Production orders were joined with the Gold Layer inventory status table.The quantity required by each production order was compared against required by each production order was compared against the available stock quantity.Production orders was classified as unfulfilled when Required Quantity > Current Stock .The SQL logic used to identify production shortages is shown in Figure 5.2.4.2.

```sql
CREATE OR ALTER VIEW dbo.gold_unfulfilled_production
AS

SELECT

p.production_order_id,
p.finished_product,
p.material_id,

p.quantity_consumed,

f.current_stock,

(p.quantity_consumed - f.current_stock)
AS shortfall

FROM ppg_silver.dbo.silver_production_orders p

INNER JOIN dbo.fact_inventory_status f
ON p.material_id = f.material_id

WHERE p.quantity_consumed > f.current_stock;
```

Figure 5.2.4.2: Gold Q6 Unfulfilled Production Query Script

Q7 At-risk sales orders tracing query

This query aimed to identify open customer sales orders that may be affected by materials classified as stockout risk. The analysis mapped relationships between customer orders, production orders, and inventory materials by linking sales orders to production orders through the finished product, and then connecting production orders to the inventory status table. Materials flagged as stockout risk were used to determine which customer orders could be impacted.

This relational flow demonstrates how customer orders are traced through production and inventory layers to identify potential risks from insufficient material availability.showing the sequence:
Customer Sales Orders → Finished Product → Production Orders → Material → Stockout Risk.

The SQL view presented in Figure 5.2.4.3 detects production orders where the required material quantity exceeds available stock.It lays the foundation for tracking high-risk sales orders related to overstock materials, and helps to achieve proactive inventory management and customer order management.

```sql
CREATE OR ALTER VIEW dbo.gold_unfulfilled_production
AS

SELECT

p.production_order_id,
p.finished_product,
p.material_id,

p.quantity_consumed,

f.current_stock,

(p.quantity_consumed - f.current_stock)
AS shortfall

FROM ppg_silver.dbo.silver_production_orders p

INNER JOIN dbo.fact_inventory_status f
ON p.material_id = f.material_id

WHERE p.quantity_consumed > f.current_stock;
```

Figure 5.2.4.3:Unfulfilled Production Analysis in At-Risk Sales Orders Tracing Script

5.2.5 Gold Layer Export – Conversion to Parquet Format

Once we finish cleaning and enhancing the data in the Gold layer, the CETAS (Create External Table As Select) method to export our final datasets. This approach is a highly effective way to pull processed data from SQL views into external storage, all while keeping the data perfectly intact and making our future analysis run much faster.

External tables were also created for each of the Gold Layer datasets: fact_inventory_status, gold_inventory_extended, gold_unfulfilled_production, and gold_sales_impact. Parquet is a format of columnar data storage and compression that benefits query performance and compression over CSV files, and these data were stored in it. This decreases the storage needs and allows rapid access to data and efficient use of resources. The standardised structure of Parquet also allows for downstream analysis processes like calculation of business metrics, creation of KPIs, reporting and visualization of dashboards.

Figure 5.2.5.1 illustrates the CETAS process used to export the Gold layer dataset into Parquet.By referencing the data source and file format directly within the external tables, ensuring a smooth integration with analytics tools like Azure Synapse Analytics and Power BI. Ultimately, this strategy helps to optimize queries efficiency, reduce storage requirement, and build a scalable business intelligence reporting system.

```sql
CREATE EXTERNAL TABLE dbo.fact_inventory_status_parquet
WITH (
    LOCATION = 'fact_inventory_status',
    DATA_SOURCE = gold_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT *
FROM dbo.fact_inventory_status;
CREATE EXTERNAL TABLE dbo.gold_inventory_extended_parquet
WITH (
    LOCATION = 'gold_inventory_extended',
    DATA_SOURCE = gold_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT *
FROM dbo.gold_inventory_extended;
CREATE EXTERNAL TABLE dbo.gold_unfulfilled_production_parqu
WITH (
    LOCATION = 'gold_unfulfilled_production',
    DATA_SOURCE = gold_ds,
    FILE_FORMAT = parquet_ff
)
AS
SELECT *
FROM dbo.gold_unfulfilled_production;
CREATE EXTERNAL TABLE dbo.gold_sales_impact_parquet
WITH (
    LOCATION = 'gold_sales_impact',
    DATA_SOURCE = gold_ds,
    FILE_FORMAT = parquet_ff
```

Figure 5.2.5.1: CETAS Script for Gold Layer Parquet Conversion

5.3 Interfaces of System Main Functions

This section shows the interface of each main function in the system and provides a clear description of things contained within it.

5.3.1 Bronze Layer - Azure Setup & Ingestion

1. Azure Portal : Resource Group rg-ppg showing all 6 resources

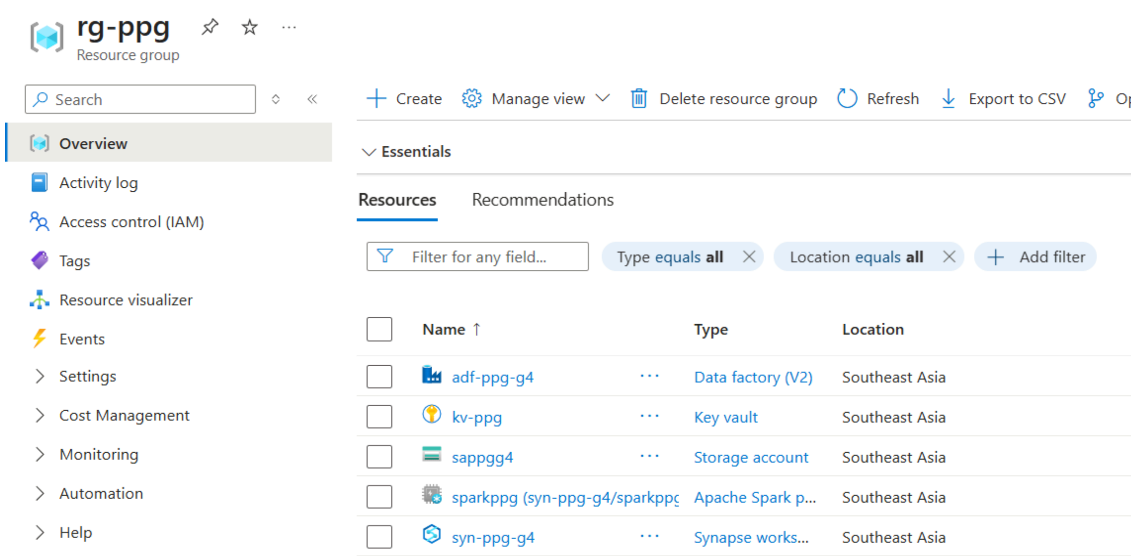


Figure 5.3.1.1 Azure Resource Group interface

Figure 5.3.1.1 shows the Azure Portal view of the resource group rg-ppg, which contains all six Azure resources used in this project. The resources include the Data Factory (adf-ppg-g4), Key Vault (kv-ppg), Storage Account (sappgg4), Apache Spark pool (sparkppg), and Synapse workspace (syn-ppg-g4). All resources are deployed in the Southeast Asia region. Grouping them under one resource group makes it easier to manage, monitor, and control access to the whole system

2. Storage Account : 4 containers listed (bronze, silver, gold, synapse-fs)

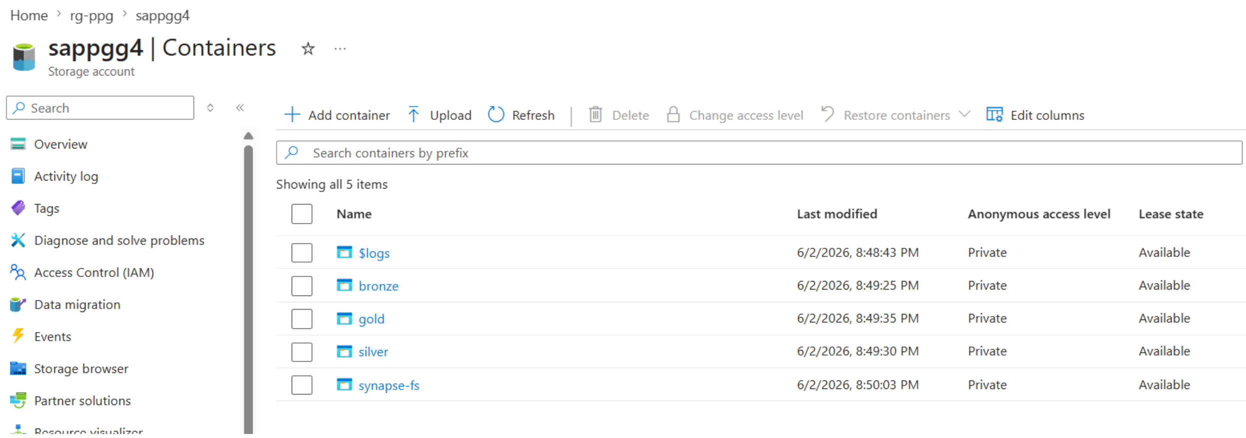


Figure 5.3.1.2 Storage Account interface

Figure 5.3.1.2 shows the storage account sappgg4 with all five containers listed: $logs, bronze, gold, silver, and synapse-fs. Each container is set to private access for security. The bronze, silver, and gold containers represent the three Medallion Architecture layers, while synapse-fs is used by the Synapse workspace. This structure ensures that data at each stage of the pipeline is stored separately and is easy to organise.

3. Bronze container : 6 subfolders with CSV files uploaded

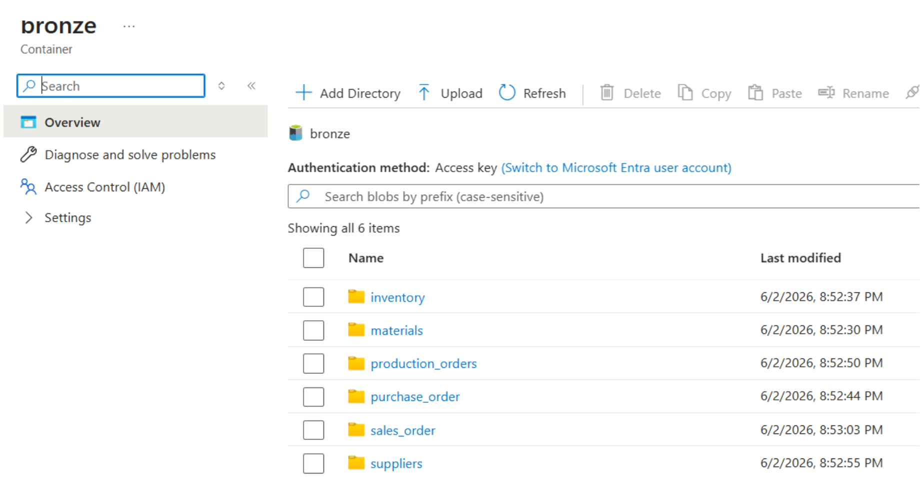


Figure 5.3.1.3 Storage Account Bronze layer interface

Figure 5.3.1.3 shows the inside of the bronze container, which has six subfolders, one for each source dataset: inventory, materials, production_orders, purchase_order, sales_order, and suppliers. Each subfolder stores its own raw CSV file. This folder structure keeps the raw data clearly separated by dataset and makes it easier to reference each file in later pipeline steps.

4. ADF Studio : pl_bronze_ingest pipeline canvas with 6 Copy activities

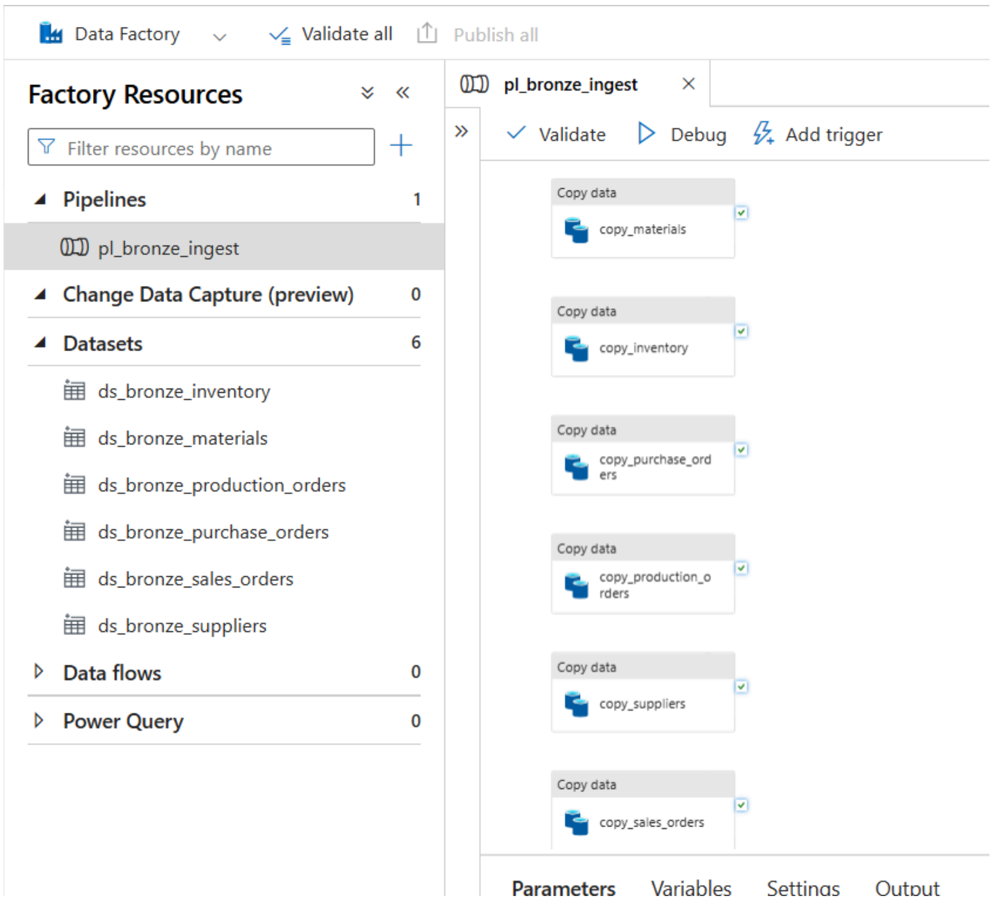
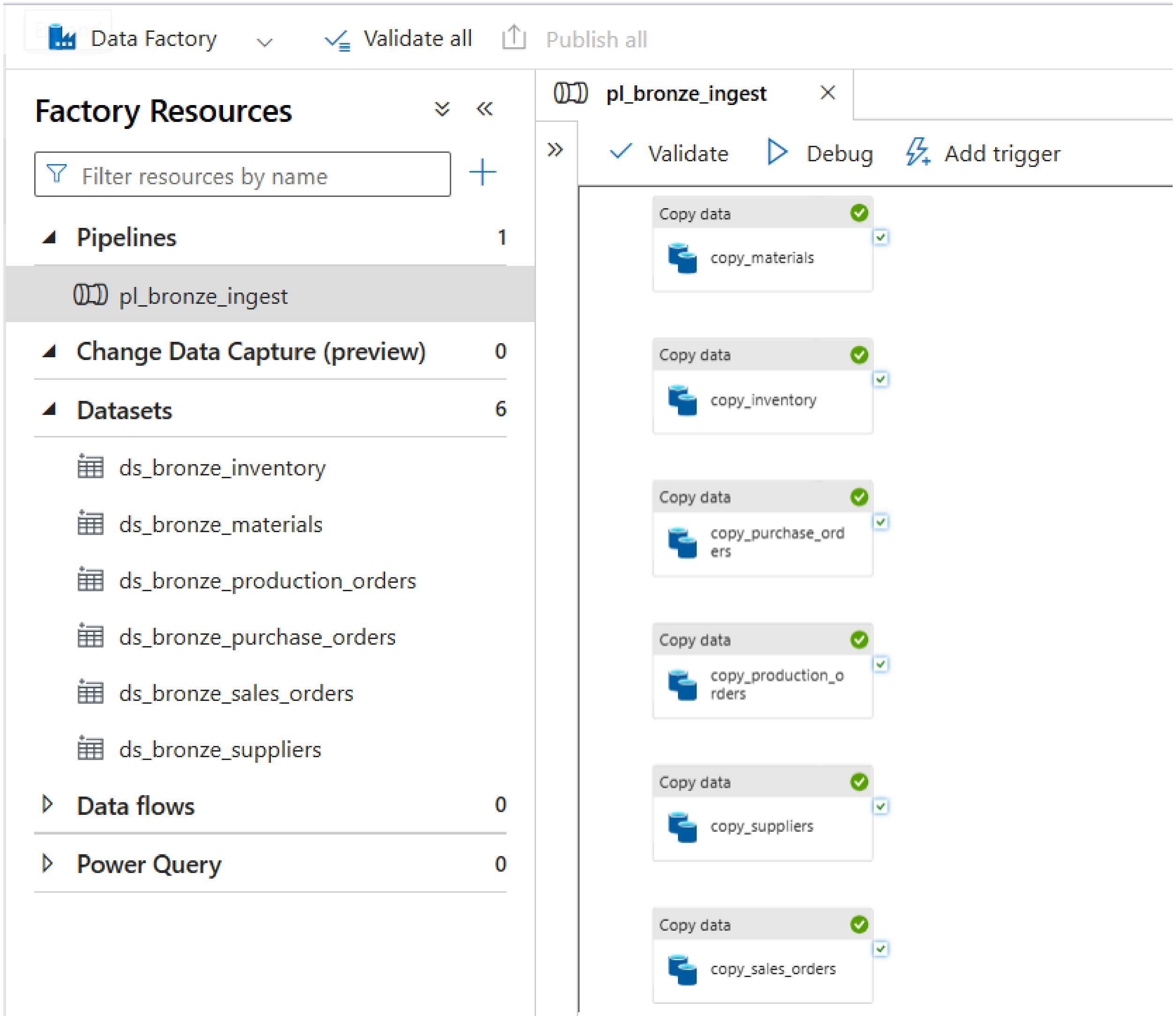


Figure 5.3.1.4 Azure Data Factory interface

Figure 5.3.1.4 shows the Azure Data Factory Studio with the pl_bronze_ingest pipeline open on the canvas. The pipeline contains six Copy Data activities (copy_materials, copy_inventory, copy_purchase_orders, copy_production_orders, copy_suppliers, and copy_sales_orders) and uses six datasets (ds_bronze_inventory, ds_bronze_materials, and so on). This single pipeline is the main tool used to load all six raw CSV files into the Bronze container in one run.

5. ADF Monitor : Pipeline run showing all 6 activities green (success)



Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...
copy_suppliers	Succeeded	Copy data	6/8/2026, 2:26:28 PM	20s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_sales_orders	Succeeded	Copy data	6/8/2026, 2:26:28 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_inventory	Succeeded	Copy data	6/8/2026, 2:26:28 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_purchase_orders	Succeeded	Copy data	6/8/2026, 2:26:28 PM	19s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_materials	Succeeded	Copy data	6/8/2026, 2:26:28 PM	20s	AutoResolveIntegrationRuntime (Southeast Asia)	
copy_production_orders	Succeeded	Copy data	6/8/2026, 2:26:28 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)	

Figure 5.3.1.5 Pipeline Execution Monitoring

Figure 5.3.1.5 shows the ADF Monitor page after running the pl_bronze_ingest pipeline. All six Copy Data activities show a green Succeeded status, with run durations between 18 and 20 seconds each. This confirms that every CSV file was copied into the Bronze container without any error, proving that the ingestion process is working correctly and is ready for the next stage of the pipeline.

6. Synapse Studio : All 6 audit query results (one screenshot per query)

Audit Query 1 - Materials

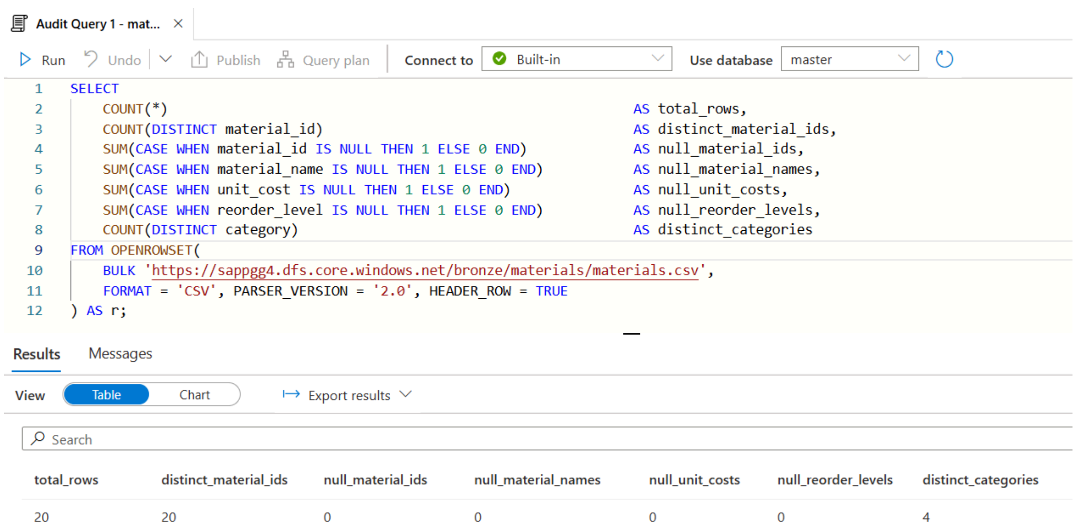


Figure 5.3.1.6 Audit Query 1

Audit Query 2 - inventory_snapshot

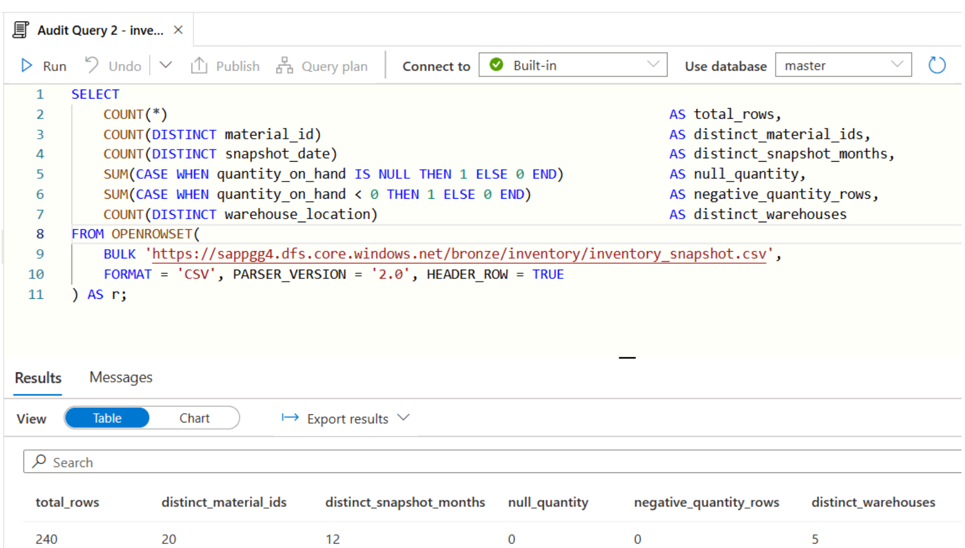
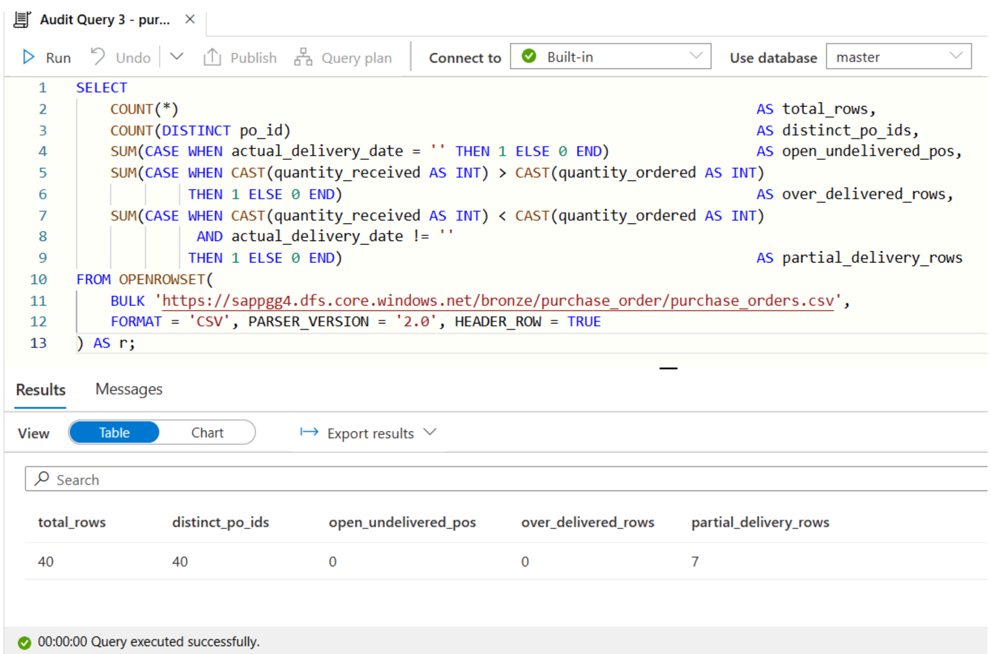


Figure 5.3.1.7 Audit Query 2

Audit Query 3 - purchase_orders

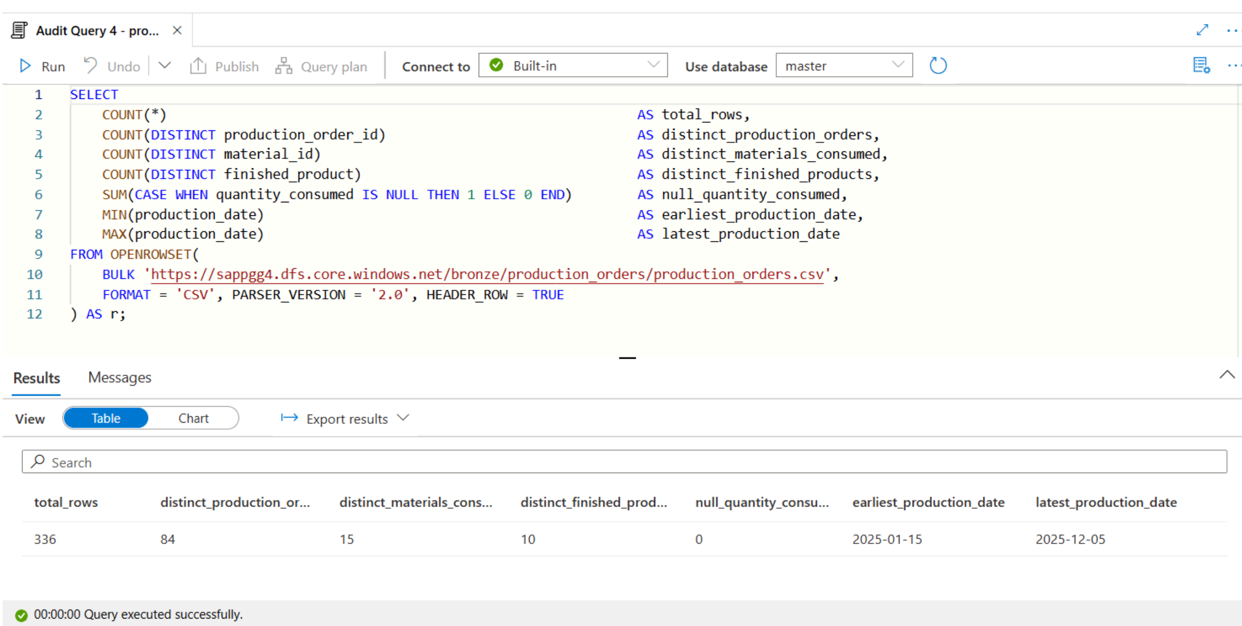



```sql
SELECT
    COUNT(*)                                                              AS total_rows,
    COUNT(DISTINCT po_id)                                                 AS distinct_po_ids,
    SUM(CASE WHEN actual_delivery_date = '' THEN 1 ELSE 0 END)            AS open_undelivered_pos,
    SUM(CASE WHEN CAST(quantity_received AS INT) > CAST(quantity_ordered AS INT)
             THEN 1 ELSE 0 END)                                           AS over_delivered_rows,
    SUM(CASE WHEN CAST(quantity_received AS INT) < CAST(quantity_ordered AS INT)
              AND actual_delivery_date != ''
             THEN 1 ELSE 0 END)                                           AS partial_delivery_rows
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/purchase_order/purchase_orders.csv',
    FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
) AS r;
```

total_rows	distinct_po_ids	open_undelivered_pos	over_delivered_rows	partial_delivery_rows
40	40	0	0	7

Figure 5.3.1.8 Audit Query 3

Audit Query 4 - production_orders




```sql
SELECT
    COUNT(*)                                                    AS total_rows,
    COUNT(DISTINCT production_order_id)                         AS distinct_production_orders,
    COUNT(DISTINCT material_id)                                 AS distinct_materials_consumed,
    COUNT(DISTINCT finished_product)                            AS distinct_finished_products,
    SUM(CASE WHEN quantity_consumed IS NULL THEN 1 ELSE 0 END)  AS null_quantity_consumed,
    MIN(production_date)                                        AS earliest_production_date,
    MAX(production_date)                                        AS latest_production_date
FROM OPENROWSET(
    BULK 'https://sappgg4.dfs.core.windows.net/bronze/production_orders/production_orders.csv',
    FORMAT = 'CSV', PARSER_VERSION = '2.0', HEADER_ROW = TRUE
) AS r;
```

total_rows	distinct_production_or...	distinct_materials_cons...	distinct_finished_prod...	null_quantity_consu...	earliest_production_date	latest_production_date
336	84	15	10	0	2025-01-15	2025-12-05

Figure 5.3.1.9 Audit Query 4

Audit Query 5 - suppliers

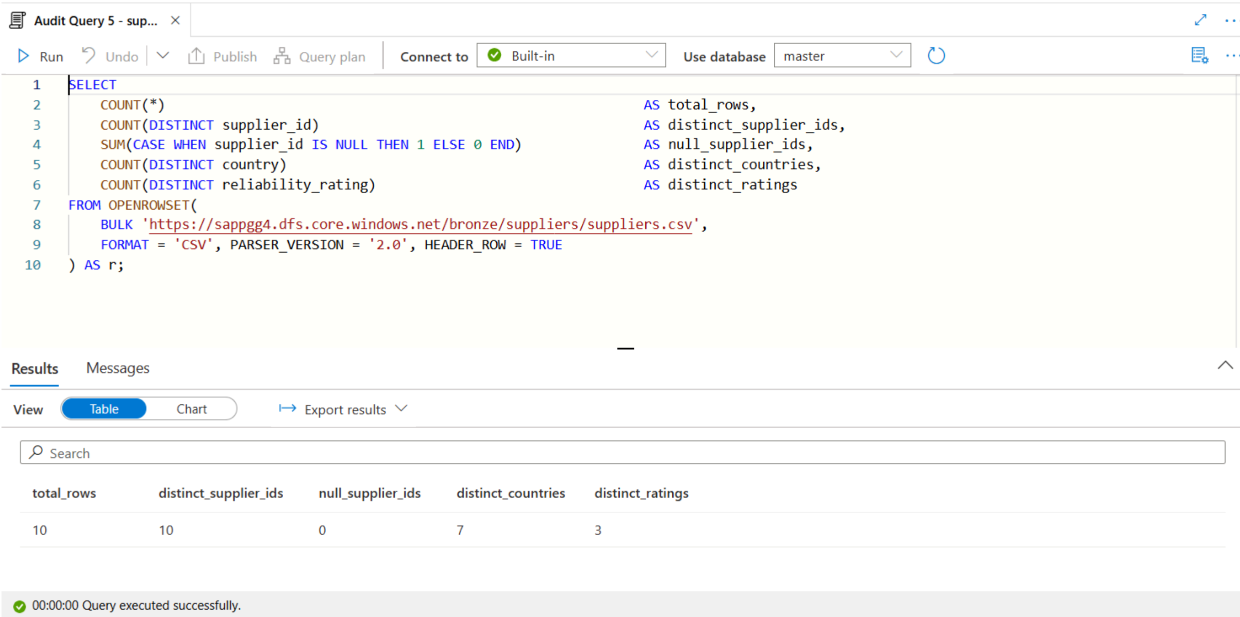


Figure 5.3.1.10 Audit Query 5

Audit Query 6 - sales_orders

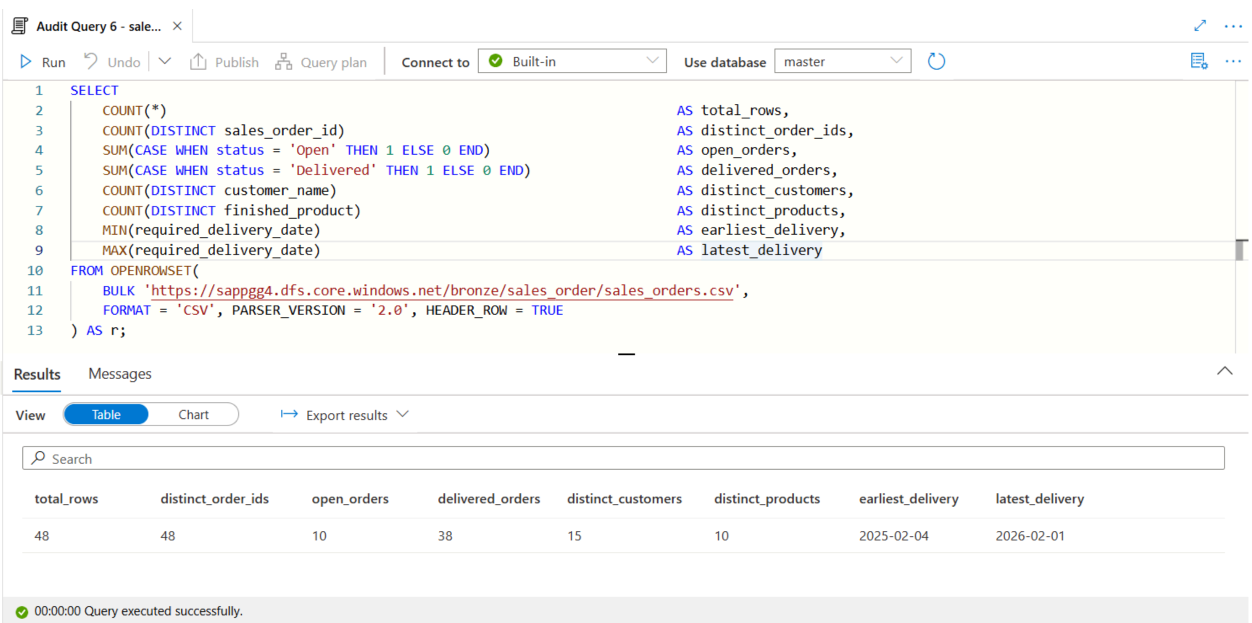


Figure 5.3.1.11 Audit Query 6

Figure 5.3.1.6 until figure 5.3.1.11 shows the six audit queries run in Synapse Studio to verify the data inside the Bronze container. Each query reads one CSV file using OPENROWSET and counts the rows, checks for null values, and inspects key columns. The results confirm: 20 materials with 4 categories, 240 inventory rows across 12 months, 40 purchase orders, 336 production orders, 10 suppliers from 7 countries, and 48 sales orders (10 Open and 38 Delivered). All results match the expected values, which proves that the Bronze Layer ingestion is complete and accurate.

5.3.2 Silver Layer Interfaces

The Azure Silver Layer was implemented and verified using Azure Synapse Studio and Azure Data Lake Storage Gen2. The transformation scripts were run in Azure Synapse Studio and the results were verified by checking that the transformed datasets were created and stored in Azure Storage.

Once the scripts in the CETAS were run successfully, six folders were created in the Silver container. Each of the folders is a cleaned version of the corresponding Bronze Layer dataset, such as Materials, Inventory, Purchase Orders, Production Orders, Suppliers and Sales Orders.

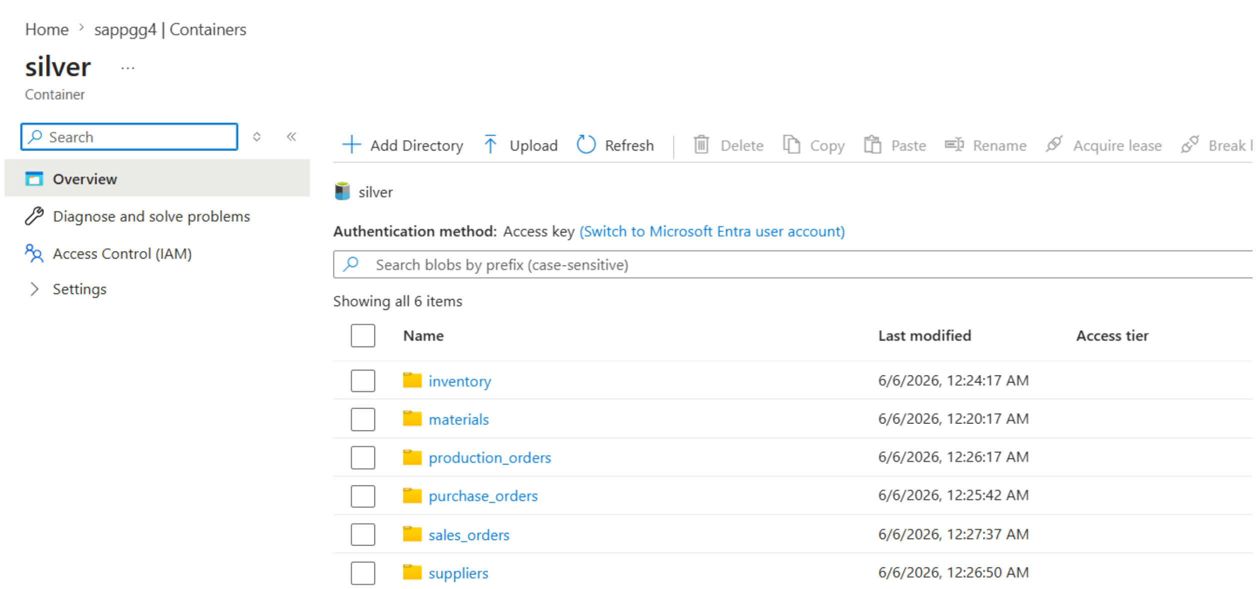


Figure 5.3.2.1: Silver Container Structure in Azure Data Lake Storage

The structure of the Silver container in which the transformed datasets are stored is shown in figure 5.3.2.1. The separation of the datasets in different folders facilitates data management and processing activities in the future. Based on figure 5.3.2.2, the transformed datasets were created as Parquet files within each folder. The cleaned and standardized data generated in the Silver Layer transformation process are contained in these files.

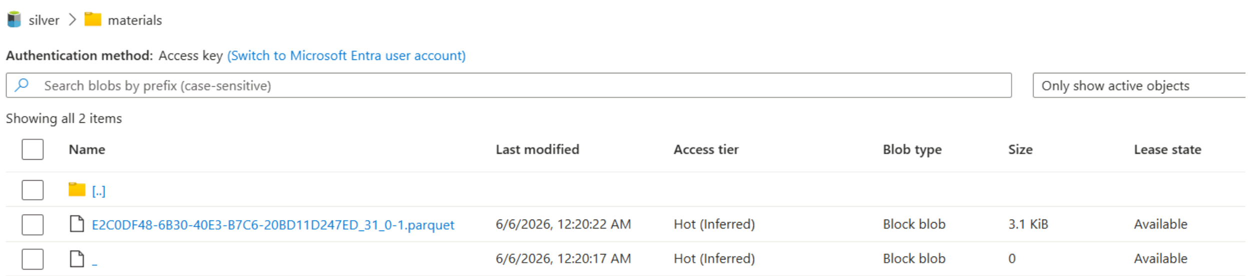


Figure 5.3.2.2: Example of Generated Parquet File in the Silver Layer

The successful creation of the Parquet files indicates that the data transformation process was completed successfully. These datasets are then used as the primary input source for the Gold Layer analytical processing stage.

5.3.3 Gold RA/MC Interfaces

This section displays the outcomes of the Gold Layer business logic implemented in Azure Synapse Analytics. To tackle business challenges Q1 to Q5, the interfaces display the results of the financial provisioning assessment and inventory risk categorization. The view fact_inventory_status, which contains the ultimate inventory status data generated after implementing the Recoverable Assets (RA) and Magna Carta (MC) business rules, yielded the query results. Furthermore, the Azure Data Lake Storage structure of the Gold Layer dataset is illustrated.

Q1 Query Result (materials below reorder level)

The materials that have reached a critical inventory state are listed in the reorder query result below. In order to avoid production disruptions, this interface helps users identify goods that require replenishment planning.

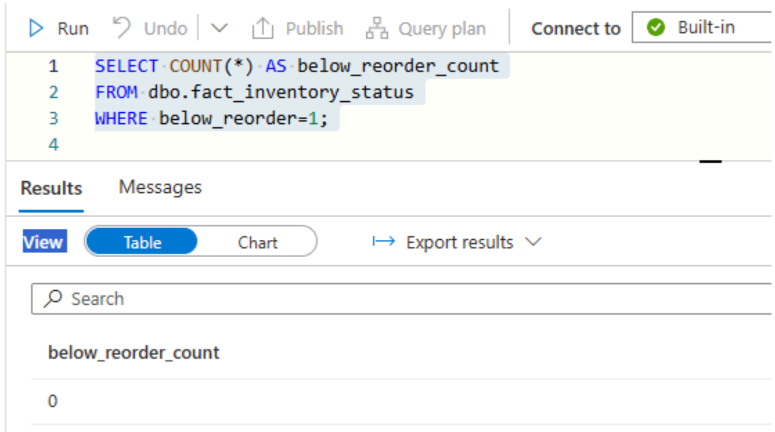


Figure 5.3.3.1: Query Result for Materials Below Reorder Level

The query results for items below the ordering threshold are displayed in Figure 5.3.3.1. The result provides information about the affected commodities and the state of their inventory. These records can be examined by users to determine which materials need immediate attention.

Q2 Query Result (Recoverable Assets list)

Items identified as surplus inventory based on consumption over the last 12 months are displayed on the Recoverable Assets portal.

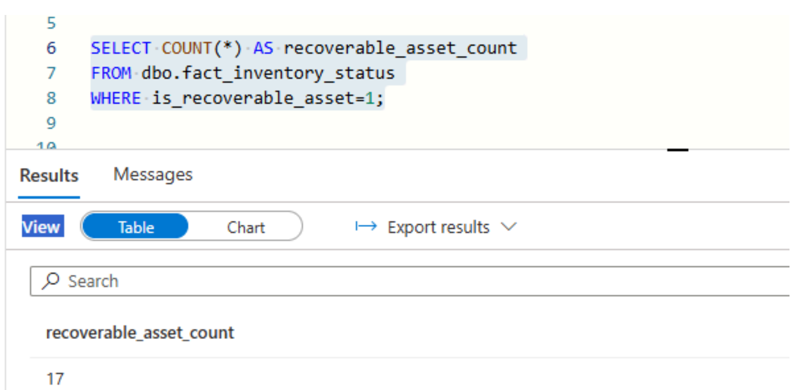


Figure 5.3.3.2: Query Result for Recoverable Assets

The catalog of materials classified as Recoverable Assets is shown in Figure 5.3.3.2. The interface helps management evaluate possible reuse, redistribution, or consumption planning tactics and provides insights into surplus inventory problems.

Q3 Query Result (MC Dead Stock materials)

Materials that have not been used in accordance with the predefined business standards are displayed on the MC Dead Stock interface.

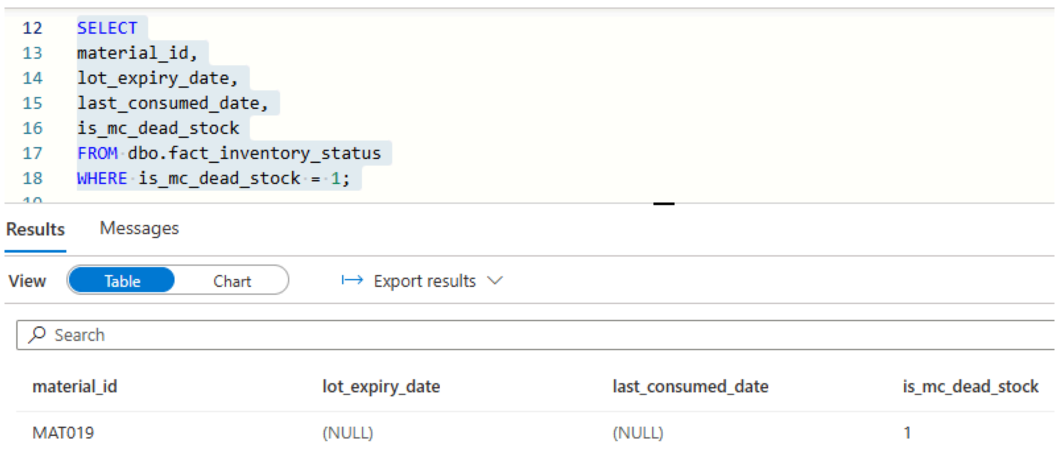


Figure 5.3.3.3: Query Result for MC Dead Stock Materials

The results of the query for Magna Carta Dead Stock items are displayed in Figure 5.3.3. Through the interface, users can identify out-of-date inventory and evaluate the potential financial impact of extra stock.

Q4 Query Result (MC Expired materials)

Information on inventory goods that are past their expiration date is provided by the expired materials interface.

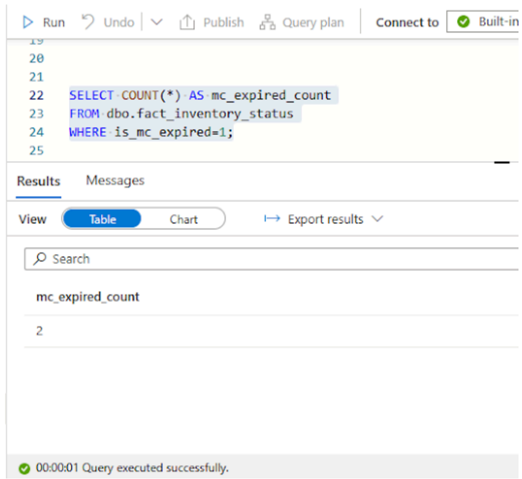


Figure 5.3.3.4: Query Result for MC Expired Materials

Figure 5.3.3.4 displays the analysis results of the expired material. Users can track expired inventory and make judgments about disposal or financial adjustments with the help of the interface.

Q5 Query Result (total provision amount)

The total financial provision required for identified obsolete inventory is described in the provision analysis interface. This outcome helps with risk assessment and financial reporting.

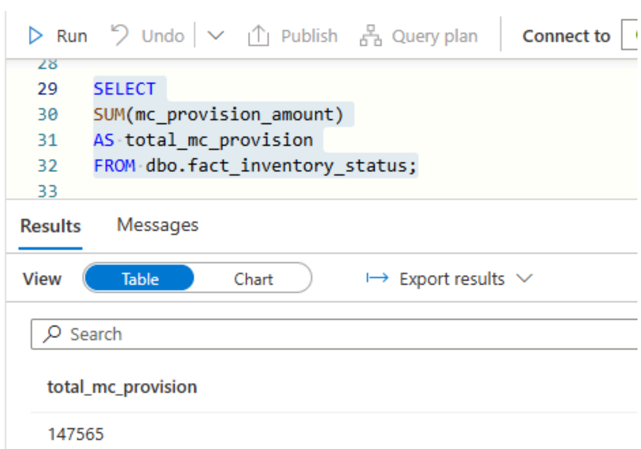


Figure 5.3.3.5: Query Result for Total MC Provision Amount

The calculated provision amount generated by the Gold Layer is shown in Figure 5.3.3.5. The result provides an overview of potential financial hazards associated with out-of-date materials and Magna Carta Dead Stock.

Azure Portal – Gold Container

The Azure Portal's Gold Container contains the most recent refined analytics-ready data. Organized outputs are stored in the Gold Layer once all transformations and business logic computations are finished.

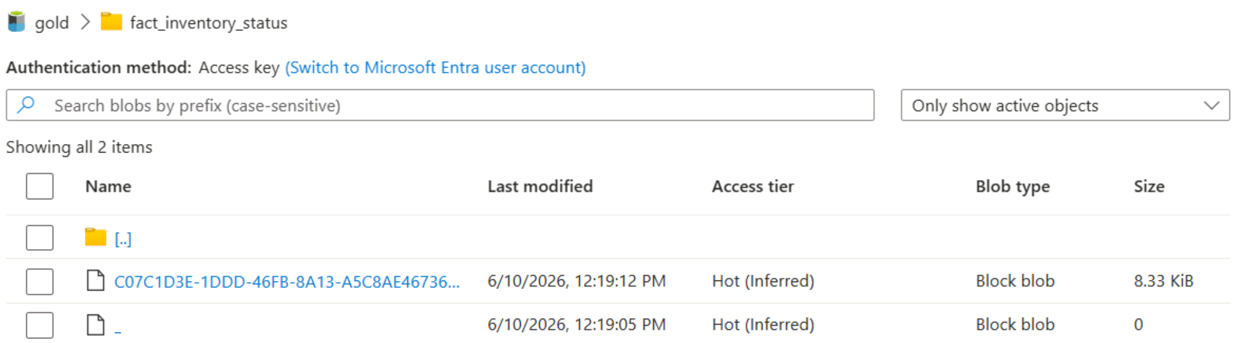


Figure 5.3.3.6: Gold Layer Parquet Files Stored in Azure Data Lake Storage

The Azure Portal's Gold Container architecture is shown in Figure 5.3.3.6. In addition to storing the finished datasets produced by the pipeline, the container offers the crucial data source for analytical and visualization tasks. This guarantees that business users can obtain dependable and organized inventory data through reporting interfaces.

5.3.4 Gold Sales Impact Interfaces

D1 Data Extension Implementation

The output shows the implementation of data extension that fills the inventory data through the attribute lot_expiry_date in the gold_inventory_extended view. The generated expiration date can be used for inventory analysis based on expiration and can be used to implement business rules, such as Magna Carta Dead Stock (Q3), Magna Carta Expired (Q4), and MC Provision (Q5) business rules.

```sql
SELECT TOP 10 *
FROM dbo.gold_inventory_extended;
```

Results Messages

View Table Chart Export results

Search

snapshot_date	material_id	quantity_on_hand	warehouse_loc...	lot_expiry_date
2025-01-31	MAT001	2000	WH-A	2025-06-01
2025-01-31	MAT002	600	WH-B	2026-12-31
2025-01-31	MAT003	1360	WH-C	2026-12-31
2025-01-31	MAT004	360	WH-D	2026-12-31
2025-01-31	MAT005	400	WH-E	2025-06-01
2025-01-31	MAT006	2430	WH-A	2026-12-31
2025-01-31	MAT007	1950	WH-B	2026-12-31
2025-01-31	MAT008	900	WH-C	2026-12-31
2025-01-31	MAT009	1200	WH-D	2026-12-31
2025-01-31	MAT010	800	WH-E	2026-12-31

Figure 5.3.4.1: Output of Gold Inventory Extended View

Figure 5.3.4.1 shows the results of the changes made after adding the lot_expiry_date_date attribute to the gold_inventory_inventory table. This view contains the modified results after adding the lot_expiry_date_date attribute to the gold_inventory table.Materials MAT001 and MAT005 have been assigned expiration dates , materials MAT016 to MAT020 have NULL values and the rest of the materials have a valid expiration date according to the predefined business rules

Q6 Unfulfilled Production Orders Analysis

The analysis is carried out using the gold_unfulfilled_production_production view. This view shows the material requirements in the production order and compares it with the current inventory level to highlight the production orders that cannot be completed due to insufficient inventory.

```
SELECT * FROM dbo.gold_unfulfilled_production
```

Results Messages

View Table Chart Export results

Search

production_order_id	finished_product	material_id	quantity_consumed	current_stock	shortfall

Figure 5.3.4.2: Output of the gold_unfulfilled_production View

The result of the Gold_unfulfilled_production view is shown in Figure 5.3.4.2. No record found, which means that all production orders can be completed using the current inventory and no production interruption is detected.

Q7 At-Risk Sales Orders and Affected Customers

The gold_sales_impact view is used for Q7 analysis. This view traces back from production data and sales order data to shortage-risk materials to identify customer sales orders and customers affected by shortages.

```
SELECT *
FROM dbo.gold_sales_impact;
```

Results Messages

View Table Chart Export results

Search

sales_order_id	customer_name	finished_product	quantity_order...	required_deliv...	material_id	material_name	current_stock	stockout_risk
SO-045	Maybank Facilit...	Marine Blue Co...	275	2026-01-24	MAT008	Epoxy Resin	1420	1
SO-047	YTL Hardware ...	Epoxy Floor Co...	385	2025-12-31	MAT008	Epoxy Resin	1420	1

Figure 5.3.4.3: Output of the gold_sales_impact View

The Gold_sales_impact view is shown in Figure 5.3.4.3. Analysis shows that two sales orders were affected which are SO-045 (Maybank facility management) and SO-047 (YTL hardware). The affected customers are Maybank Facilities Management and YTL Hardware. If the identified shortage persists, these customers may face fulfilment issues due to insufficient inventory.

Gold Layer Output Storage

The Parquet file is stored in the Gold Gold container of Azure Storage as shown in Figure 5.3.4.4. These files are the final output generated by the Gold Gold layer after applying the inventory risk management business rules and analysing and processing. The stored output includes fact_inventory_status, which includes inventory risk indicators and business indicators.In addition, D1 data extensions called gold_inventory_extended, unfinished production order analysis (Q6), and high-risk sales orders and affected customer analysis (Q7) have been added to store the results. These Parquet files have analysis functions and can be used for reports and dashboard visualisation in Power BI.

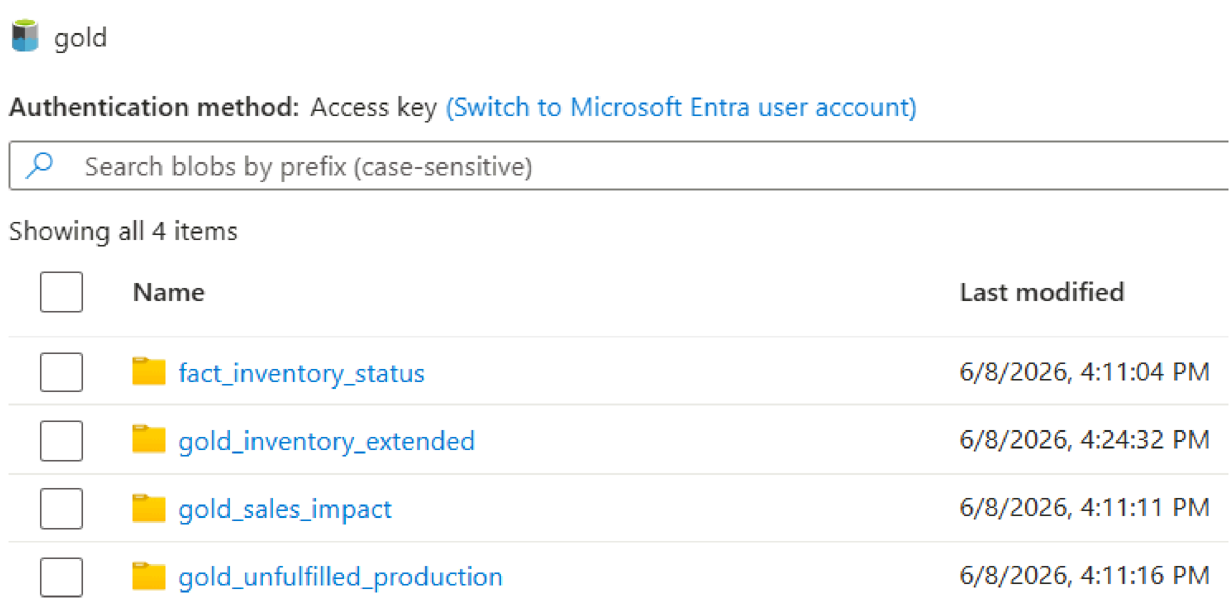


Figure 5.3.4.4: Gold Layer Output Files Stored in Azure Storage

5.4 Testing

This section discussed the testing process that has been conducted to verify the system performance and making sure everything is working as expected.

5.4.1 Bronze Layer Testing

This part shows the specific test case that was conducted to verify whether the ingestion of raw data into the bronze layer is successful or not. Table 5.4.1.1 are the details of the expected results and actual outcome for each data source.

Test ID	Test Case	Expected Result	Succeed / Failed
A001	materials.csv ingested	20 rows, 0 null material_ids, 4 categories	Pass
A002	inventory_snapshot.csv ingested	240 rows, 12 distinct months, 0 nulls	Pass
A003	purchase_orders.csv ingested	40 rows, 1 open undelivered PO (PO-002)	Pass
A004	production_orders.csv ingested	336 rows, 10 finished products, 0 null qty	Pass
A005	suppliers.csv ingested	10 rows, 0 null supplier_ids, 3 ratings	Pass
A006	sales_orders.csv ingested	48 rows, 10 Open + 38 Delivered	Pass

Table 5.4.1.1 Bronze layer Testing

5.4.2 Silver Layer Testing

Testing was conducted to verify that the transformed datasets were created properly and could be utilized in the subsequent phase of the data pipeline. Azure Synapse Serverless SQL Pool was used to run validation queries to ensure the contents of the generated Parquet files. The table 5.4.2.1 below shows the Silver Layer Testing.

Test ID	Test Case	Expected Result	Succeed /

			Failed
SL001	Materials Transformation	Cleaned Materials dataset generated successfully	Pass
SL002	Inventory Transformation	Cleaned Inventory dataset generated successfully	Pass
SL003	Purchase Orders Transformation	Cleaned Purchase Orders dataset generated successfully	Pass
SL004	Production Orders Transformation	Cleaned Production Orders dataset generated successfully	Pass
SL005	Suppliers Transformation	Cleaned Suppliers dataset generated successfully	Pass
SL006	Sales Orders Transformation	Cleaned Sales Orders dataset generated successfully	Pass
SL007	Parquet File Generation	Output stored in Silver container	Pass
SL008	Data Accessibility	Parquet files can be accessed using OPENROWSET	Pass

Table 5.4.2.1 Silver Layer Testing

5.4.3 Gold RA/MC Logic Testing

The Gold Layer contains the final refined data required for inventory risk assessment and reporting. To confirm that the Gold Layer's business rules accurately detect various inventory scenarios based on predetermined criteria, logic testing is carried out. The assessment focuses on verifying the accuracy of the Recoverable Assets (RA) and Magna Carta (MC) classification techniques, including provision calculation, dead stock identification, stock level analysis, and expiry detection.

Based on the established business rules, each test case evaluates if the system generates the anticipated results. The data processed in the Gold Layer is guaranteed to be dependable and appropriate for use in Power BI dashboard visualization and decision-making once these tests are successfully finished. The Gold Layer RA/MC logic test results are shown in Table 5.4.3.1, along with the test ID, test case description, expected results, and test status.

Test ID	Test Case	Expected Result	Succeed/ Failed
GL001	Q1 – Below Reorder	System identifies materials with current stock lower than the reorder level	Pass
GL002	Q2 – Recoverable Assets	System identifies materials where current stock exceeds total 12-month consumption	Pass
GL003	Q3 - MC Dead Stock	System identifies materials with no consumption for at least 9 months and no recorded lot expiry date	Pass
GL004	Q4 - MC Expired	System identifies materials with a lot expiry date earlier than the analysis cutoff date (31 December 2025)	Pass
GL005	Q5 - Total MC Provision	System calculates the total provision	Pass

		amount for materials classified as MC Dead Stock or MC Expired	

Table 5.4.3.1 : Gold RA/MC Testing

5.4.4 Gold Sales Impact Testing

The business logic of the golden layer was tested to verify the new functions added in inventory risk analysis and sales impact analysis. The test content includes successfully generating D1 expiration date extension, identifying expired inventory and inventory ageing.The test ensures the correct input of records, the verification of the fulfilment of production orders, and the identification of shortage risks and customer sales orders. In addition, the affected customers are accurately identified, and all golden layer output results are generated and stored correctly. All test cases have passed, as shown in Table 5.4.4.1. The business rules and analysis output of the golden layer have been correctly implemented and derived, achieving the expected results.

Test ID	Test Case	Expected Result	Succeed / Failed
GS001	D1 expiry date extension generated	240 inventory records processed; 180 records assigned valid lot_expiry_date values	Pass
GS002	Expired inventory validation	Expired inventory records successfully identified	Pass
GS003	Q6 unfulfilled production analysis	Production orders successfully validated against inventory stock levels; no production shortfalls identified	Pass
GS004	Q7 at-risk sales order analysis	2 impacted sales orders identified (SO-045, SO-047)	Pass

GS005	Customer impact identification	2 affected customers identified	Pass
GS006	Gold Layer storage validation	Gold Layer output folders successfully created	Pass

Table 5.4.4.1: Gold Sales Impact Testing

5.4.5 Analytics Engine - SQL Execution of the 7 Business Questions

01_Database_Setup

Run | Undo | Publish | Query plan | Connect to | Built-in

```sql
CREATE DATABASE ppg_curated;
GO
USE ppg_curated;
GO
-- External Data Source pointing to Gold container
CREATE EXTERNAL DATA SOURCE gold_ds WITH (
    LOCATION = 'https://sappgg4.dfs.core.windows.net/gold'
);

CREATE EXTERNAL FILE FORMAT csv_ff WITH (
    FORMAT_TYPE = DELIMITEDTEXT,
    FORMAT_OPTIONS (
        FIELD_TERMINATOR = ',',
        STRING_DELIMITER = '"',
        FIRSTROW = 2 -- untuk skip column header
    )
);

-- Parquet file format
CREATE EXTERNAL FILE FORMAT parquet_ff WITH (FORMAT_TYPE = PARQUET);
```

Figure 5.4.5.1: Curated Database and External Storage Configuration

To transition the processed datasets from the Silver layer into highly optimized analytical models, a dimensional Star Schema was engineered in the Gold curated layer. This architecture organizes data into centralized Fact tables capturing transactional metrics surrounded by descriptive Dimension tables to optimize query retrieval speeds.

A dedicated database catalog named ppg_curated is created, and an external data source gold_ds is configured to point directly to the Azure Data Lake Storage container endpoint at https://sappgg4.dfs.core.windows.net/gold. External file formats for both delimited text (csv_ff) and columnar data (parquet_ff) are explicitly defined to enable direct database queries over the underlying storage files.

Q1 - Below Safe Reorder Levels

Running this query returned an empty result set across all columns (material_id, material_name, category, current_stock, reorder_level, shortfall). This confirms that there are currently zero inventory items breaching active safety buffer thresholds.

Search

material_id	material_name	category	current_stock	reorder_level	shortfall

Figure 5.4.5.2: Output of Q1 Below Safe Reorder Levels

Q2 - Recoverable Assets

The query successfully flagged 17 overstocked items across the facility. The total capital locked in excess inventory equals $1,735,697.50$, with major concentrations identified in high-value components such as MAT009 (Polyurethane Resin, $418,992.00$ overstock value) and MAT007 (Acrylic Resin, $215,339.00$ overstock value).

	A
1	material_id,material_name,category,current_stock,consumption_12m,excess_stock,unit_cost,excess_value_usd
2	MAT009,Polyurethane Resin,Resin,37410,0,37410,11.2,418992
3	MAT007,Acrylic Resin,Resin,46680,6050,40630,5.3,215339
4	MAT010,Vinyl Resin,Resin,25200,0,25200,7.6,191520
5	MAT017,Metal Can 4L,Packaging,127060,5640,121420,1.4,169988
6	MAT016,Metal Can 1L,Packaging,197810,9840,187970,0.85,159774.5
7	MAT012,Toluene,Solvent,76960,1080,75880,1.6,121408
8	MAT014,Acetone,Solvent,48000,0,48000,2.1,100800
9	MAT011,Xylene,Solvent,58170,7705,50465,1.8,90837
10	MAT020,Plastic Pail 5L,Packaging,56240,2800,53440,1.1,58784.00000000001
11	MAT015,Butanol,Solvent,24030,1080,22950,2.4,55080
12	MAT019,Cardboard Box,Packaging,102000,0,102000,0.45,45900
13	MAT005,Chrome Yellow,Pigment,5300,0,5300,6.4,33920
14	MAT018,Plastic Drum 20L,Packaging,15315,5040,10275,3.2,32880

Figure 5.4.5.3: Output of Q2 Recoverable Assets Analysis

Q3 - MC Dead Stocks

Execution defines a severe stagnation event for item MAT019 (Cardbox) within the Packaging category. The ledger is recorded as current_stock of 102,000 units with completely unresolved consumption logs (NULL), triggering an automated dead stock risk provision of 45,900.

material_id	material_name	category	current_stock	last_consumed...	months_since_l...	mc_provision_...
MAT019	Cardboard Box	Packaging	102000	(NULL)	(NULL)	45900

Figure 5.4.5.4: Output of Q3 Magna Carta Dead Stock Analysis

Figure 5.4.5.4 shows the output of the Q3 Magna Carta Dead Stock analysis. The query identified material MAT019 (Cardbox) as dead stock because it had no recorded consumption activity for more than nine months and did not require expiry tracking. As a result, the material was flagged for inventory risk monitoring and provision calculation.

Q4 - MC Expired

Based on Figure 5.4.5.4, the query detects two expired items under the Pigment category:

- MAT001 (Titanium Dioxide): 7,970 units in stock at a unit_cost of 8.5, generating an mc_provision_amount liability of 67,745.
- MAT005 (Chrome Yellow): 5,300 units in stock at a unit_cost of 6.4, generating an mc_provision_amount liability of 33,920.

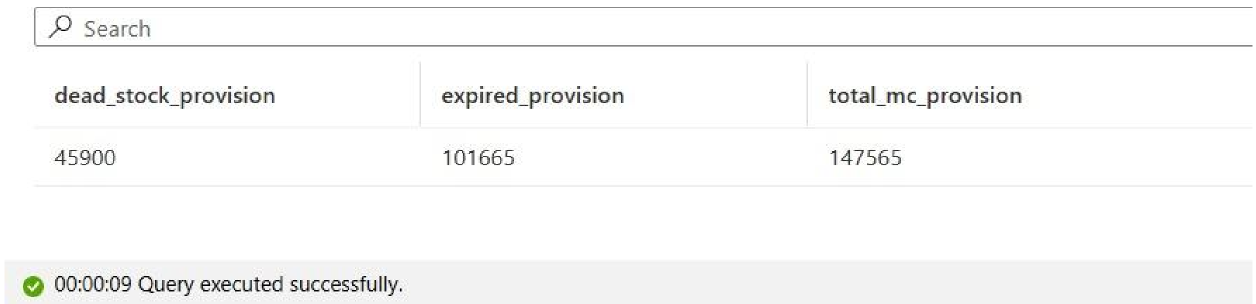
material_id	material_name	category	current_stock	unit_cost	mc_provision_amount
MAT001	Titanium Dioxide	Pigment	7970	8.5	67745
MAT005	Chrome Yellow	Pigment	5300	6.4	33920

00:00:09 Query executed successfully.

Figure 5.4.5.5: Output of Q4 Magna Carta Expired Materials Analysis

Q5 - Total MC Provision

Figure 5.4.5.6 shows the output of the Q5 Total Magna Carta (MC) Provision analysis. The query calculated a total provision amount of 147,565, consisting of provisions for both dead stock and expired materials. This value represents the total inventory write-down required based on the defined business rules.



Search

dead_stock_provision	expired_provision	total_mc_provision
45900	101665	147565

00:00:09 Query executed successfully.

Figure 5.4.5.6: Output of Q5 Total MC Provision Analysis

Q6 - Unfulfilled Production

identifies production orders that cannot be fulfilled due to insufficient inventory by comparing material requirements against available stock levels. This analysis helps detect potential production disruptions and supports proactive inventory planning.

unfulfilled_production_count
0

Q7 - At-Risk Orders

The Q7 analysis was implemented using the gold_sales_impact view. The purpose of this analysis is to identify customer sales orders and customers that may be affected by inventory stockout risks.

sales_order_id	customer_name	finished_product	quantity_order...	required_deliv...	material_id	material_name	current_stock
SO-045	Maybank Facilities Mgmt	Marine Blue Coating	275	2026-01-24	MAT008	Epoxy Resin	1420
SO-047	YTL Hardware Centre	Epoxy Floor Coating	385	2025-12-31	MAT008	Epoxy Resin	1420

Figure 5.4.5.8: Output of Q7 At-Risk Sales Orders and Affected Customers Analysis

Figure 5.4.5.8 shows the output of the Q7 analysis. The query identified a stockout risk associated with material MAT008 (Epoxy Resin), which has a current stock level of 1,420 units. The analysis revealed that two open sales orders, namely SO-047 from YTL Hardware Centre and SO-045 from Maybank Facilities Management, may be affected by the identified stockout risk. These orders require 385 units and 275 units respectively and are scheduled for delivery on 31 December 2025 and 24 January 2026. The result enables early identification of customer fulfilment risks and supports proactive inventory management decisions.

5.4.6 Enterprise Power BI Visual Analytics Dashboard

The Enterprise Power BI Visual Analytics Dashboard serves as the final presentation layer of the solution, transforming the analytical outputs generated in the Gold Layer into interactive and user-friendly business intelligence reports. The dashboard is organized into multiple report pages, each designed to address specific business functions including executive monitoring, inventory risk management, financial provision tracking, procurement decision-making, and customer impact analysis. This role-based structure enables stakeholders to efficiently access relevant insights, monitor key performance indicators (KPIs), and support data-driven decision-making across different operational areas. Table 5.4.6.1 summarizes the analytical and operational purpose of each dashboard page.

Table 5.4.6.1 *Purpose explanation of every pages in dashboard*

Report Page	**Analytical Purpose & Operational Purpose**
Executive Overview	Designed for senior management to provide a high-level operational pulse. It captures critical core statistics in prominent KPI cards: a Total MC Provision of $147.565K, an RA Materials Count of 20, a Below Reorder Count of 20, and 2 At-Risk Orders. It includes an interactive donut chart breaking down materials by classification alongside a bar chart mapping financial provision exposure by material category, showing that Pigment represents the largest liability ($102K).
Recoverable Assets Analysis	Geared toward procurement strategies and warehouse over-purchasing management (Q2). It presents a detailed Recoverable Assets List data grid tracking current stock volumes alongside an interactive Excess Value by Material treemap. This visual instantly isolates heavy capital constraints across material groups, identifying *Resin* (specifically *Polyurethane Resin* at $418.99K and *Acrylic Resin* at $215.34K) as the primary drivers of excess holding value.

Magna Carta Risk Analysis	Tailored for accounting and inventory control specialists to manage aging material blocks (Q3 & Q4). The layout splits provisions into two independent tracking tables and high-level KPI cards: Total Dead Stock Provision ($45.9K) isolating stagnant components like Cardboard Box (102,000 units), and Total Expired Provision ($101.665K) flagging passed lot lines like Chrome Yellow ($33,920) and Titanium Dioxide ($67,745).
Financial Provision Breakdown Summary	Built primarily for corporate finance auditing functions to track aggregate balance sheet liabilities (Q5). It connects individual material provision ledgers directly to a Provision Breakdown by Category waterfall chart. This layout explicitly outlines the step-by-step cost escalation from Pigment ($102K) and Packaging ($46K) up to the cumulative asset risk write-down requirement of $147,565.
Production & Customer Impact Tracing	Engineered for manufacturing schedulers and customer account managers to manage fulfillment blockages (Q6 & Q7). The view cross-references upstream material deficits directly to active downstream customer demand. It provides an operational check that flags impacted clients and their exact order volumes, identifying a total delivery threat of 660 units split between YTL Hardware Centre (385 units via SO-047) and Maybank Facilities Mgmt (275 units via SO-045) due to upstream Epoxy Resin stockouts.

5.5 Dashboard Explanation

The one-page dashboard, as illustrated in **Figure 5.5.1**, shows a full snapshot of inventory health including a detailed answer to all seven questions on the inventory status and risk metrics for this business. At first glance at the top of the dashboard, there is an overall executive view (KPI) of total MC provision in addition to the number of RA, MC Dead Stock, MC Expired products, and total number of materials at stock-out risk.

The visual layout is carefully designed to take users through key operational stages, starting with the monitoring of materials that are dropping below the reorder threshold, up to the assets that are classed as Recoverable Assets, which are then calculated for monetary value that exceeds what is deemed as excess items. In addition, it is able to recognize those materials that are no longer flowing and are identified as MC Dead Stock and MC Expired, and will combine all of the financial provision needs into a straightforward breakdown by categories. Last but not least, the dashboard clearly shows the movement from material shortages to affected customer orders, displaying unfilled production orders and displaying specific open sales orders with urgency-based conditional formatting.

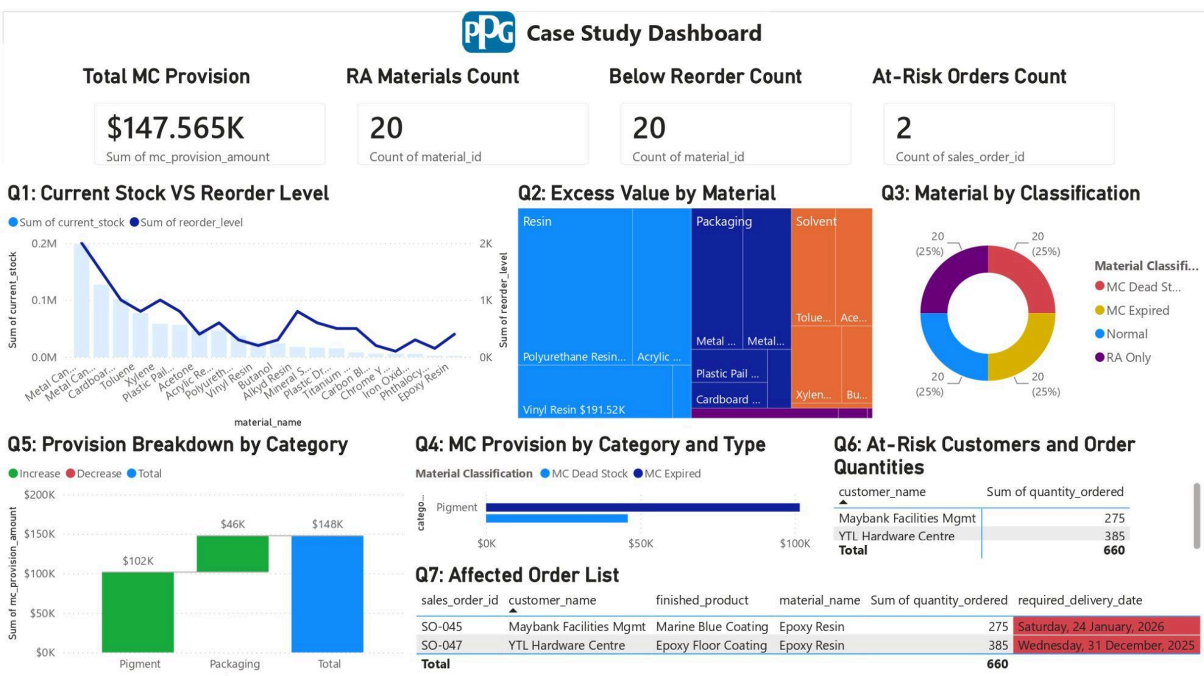


Figure 5.5.1 *Consolidated Inventory Risk Management Dashboard*

The stock level chart in **Figure 5.5.2** below shows the current stock levels for various materials against the set reorder levels. It helps the procurement team to keep a check on inventory lines continuously and immediately know which items are coming towards or going below safety stock.

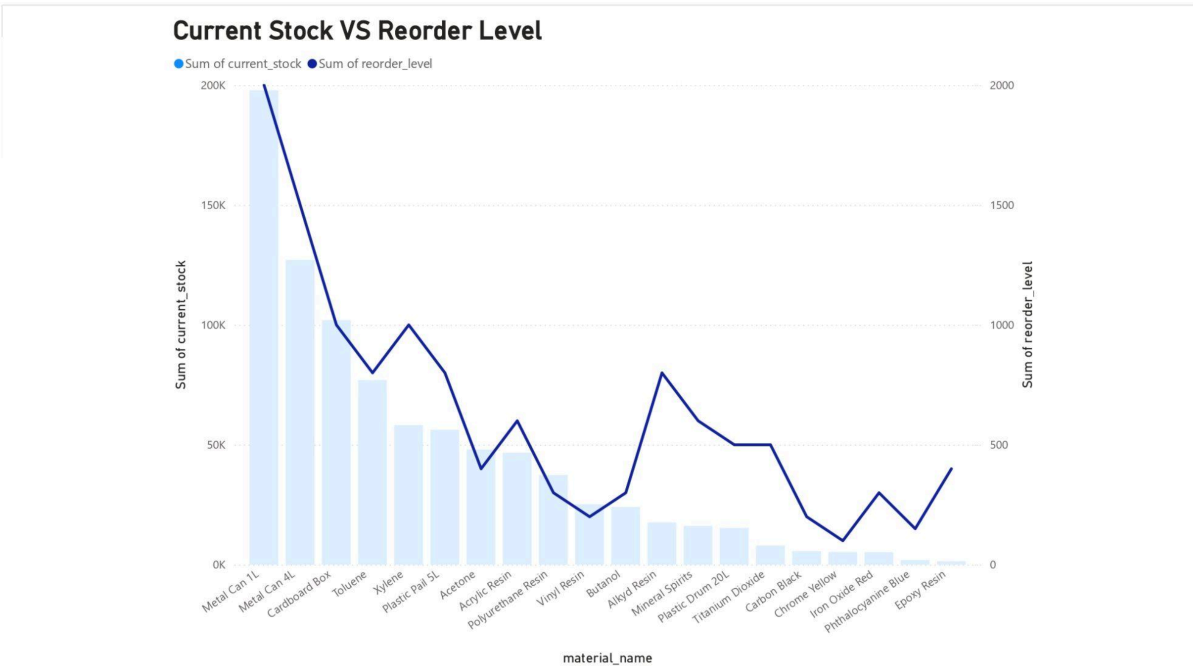


Figure 5.5.2 *Inventory Reorder Alert Chart*

The following chart below (**Figure 5.5.3**) shows the over-stocked dollar value in each of the material categories. It definitely segregates high value overstocks like Polyurethane Resin and Acrylic Resin, allowing the management team to focus on capital recovery and inventory liquidation measures.

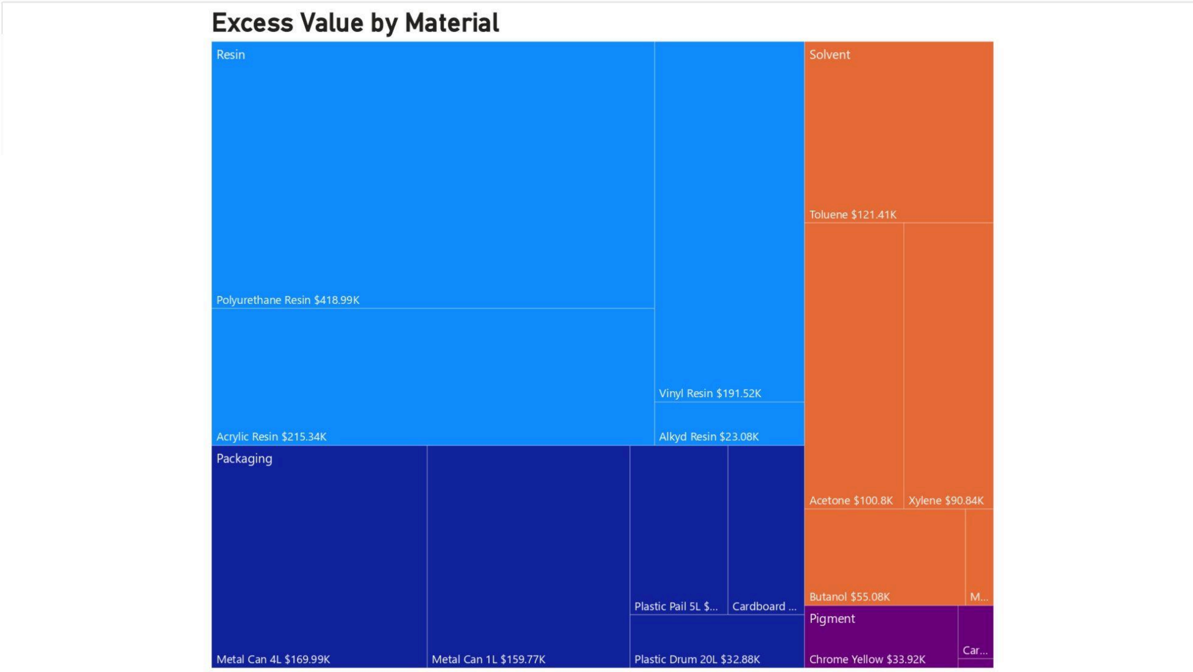


Figure 5.5.3 *Recoverable Assets Valuation Chart*

As noted in **Figure 5.5.4**, the first chart (Material by Classification) shows a fairly balanced distribution of materials across four categories: Normal, Recoverable Assets, Dead Stock, and Expired. It provides a snapshot view of the material distribution in each of the risk categories, making sure that the content is well-balanced at twenty-five percent in every category.

In the same figure below is a stacked bar chart that shows the financial write-offs by the material classification, and makes an explicit comparison between the liabilities of Dead Stock and Expired items. It shows the exact breakdown of these losses by production areas, such as Pigment and Packaging, to enable a strict accountability of costs.

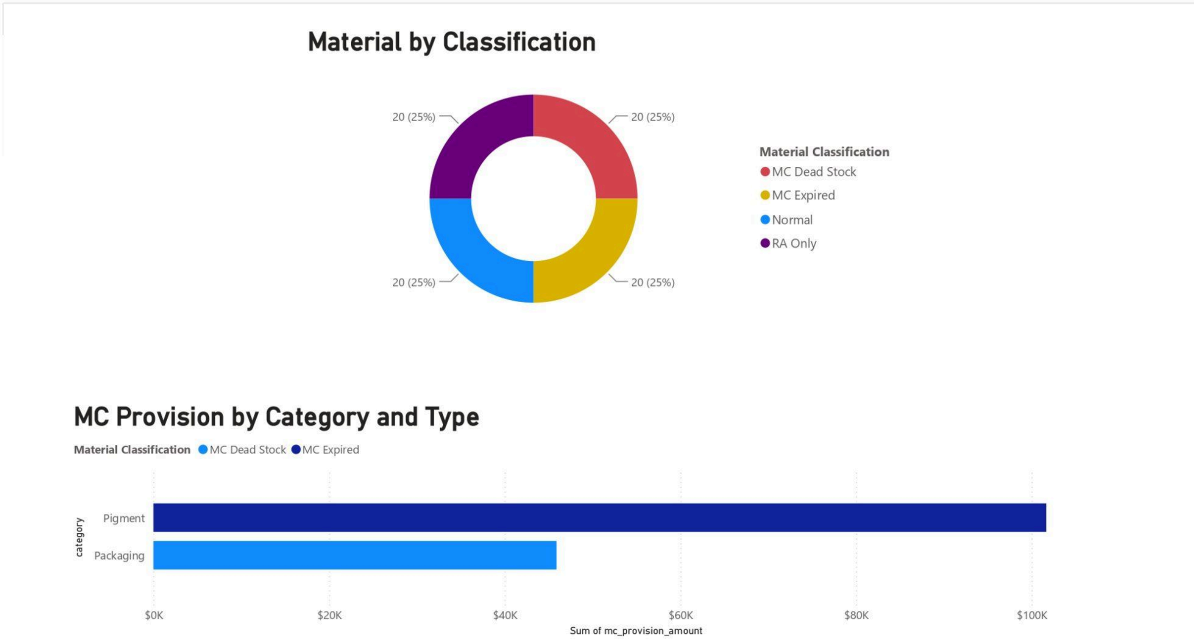


***Figure 5.5.4** Magna Carta (Dead Stock & Expired), Normal, and RA Chart*

This chart (**Figure 5.5.5**) shows the full cost of the inventory that has been written off, with a total provision costing $148K. It is able to connect the dots between the total cost, by stepping through individual financial losses from the Pigment and Packaging categories, making the total loss extremely transparent for the finance team.

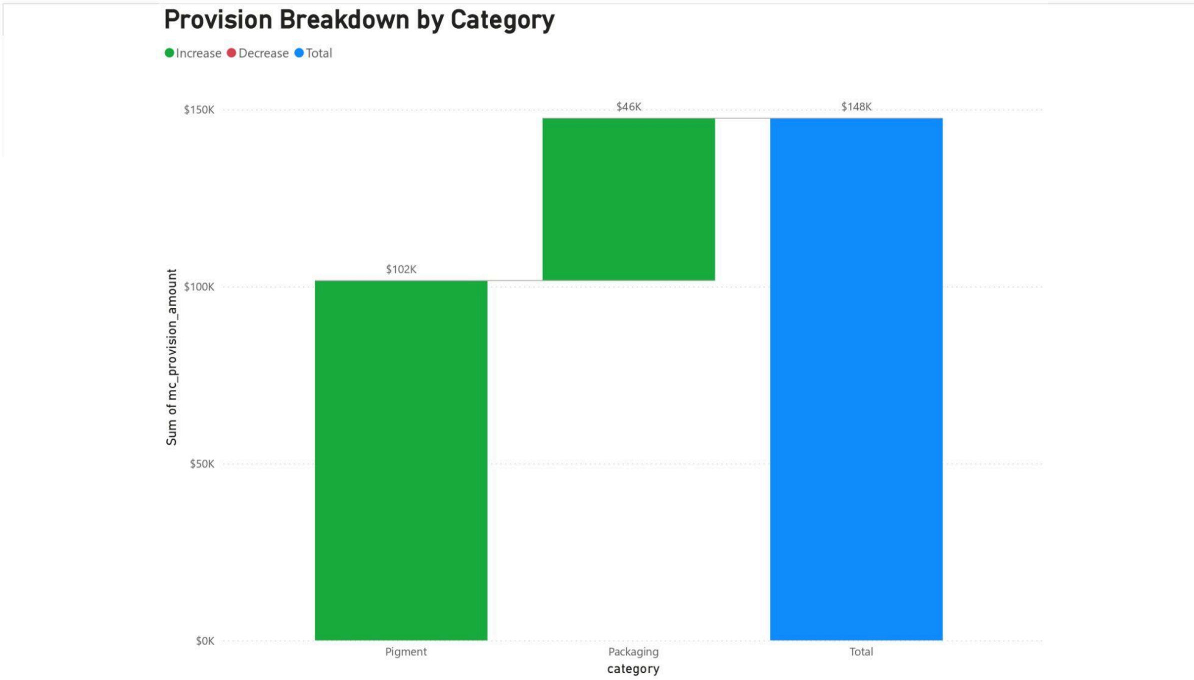


Figure 5.5.5 *Financial Provision Breakdown Chart*

The final charts in the dashboard (**Figure 5.5.6**) illustrates how clients will be affected by shortages of raw materials. From this chart, it is shown the quantities may potentially be at risk and which are the most critical customer sales such as Maybank Management and YTL Hardware Centre that the customer service and logistics teams need to prioritize in order to deliver these directly to customers.

At-Risk Customers and Order Quantities

customer_name	Sum of quantity_ordered
Maybank Facilities Mgmt	275
YTL Hardware Centre	385
Total	**660**

Affected Order List

sales_order_id	customer_name	finished_product	material_name	Sum of quantity_ordered	required_delivery_date
SO-045	Maybank Facilities Mgmt	Marine Blue Coating	Epoxy Resin	275	Saturday, 24 January, 2026
SO-047	YTL Hardware Centre	Epoxy Floor Coating	Epoxy Resin	385	Wednesday, 31 December, 2025
Total				**660**	

Figure 5.5.6 *Stockout Risk and Customer Impact Chart*

5.6 Chapter Summary

This chapter explains how the inventory risk management system for PPG industries was built using Microsoft Azure. The system uses a three-layer setup to process information. First, raw data from six sources is collected and saved in the Bronze Layer. Next, the Silver Layer cleans and organizes this data so it is easier to use. Finally, the Gold Layer applies specific business rules to the data, such as tracking expiration dates and finding out how long stock might affect production and customer orders.

After building the system, the team tested it thoroughly to make sure everything worked correctly. They checked each step of the process: making sure the raw data was collected right, verifying the data was cleaned properly, and confirming the final business calculations were accurate. All the tests passed successfully without any error. This proved that the automated system is reliable and meets all the project’s goals.

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